

BFS Traversal

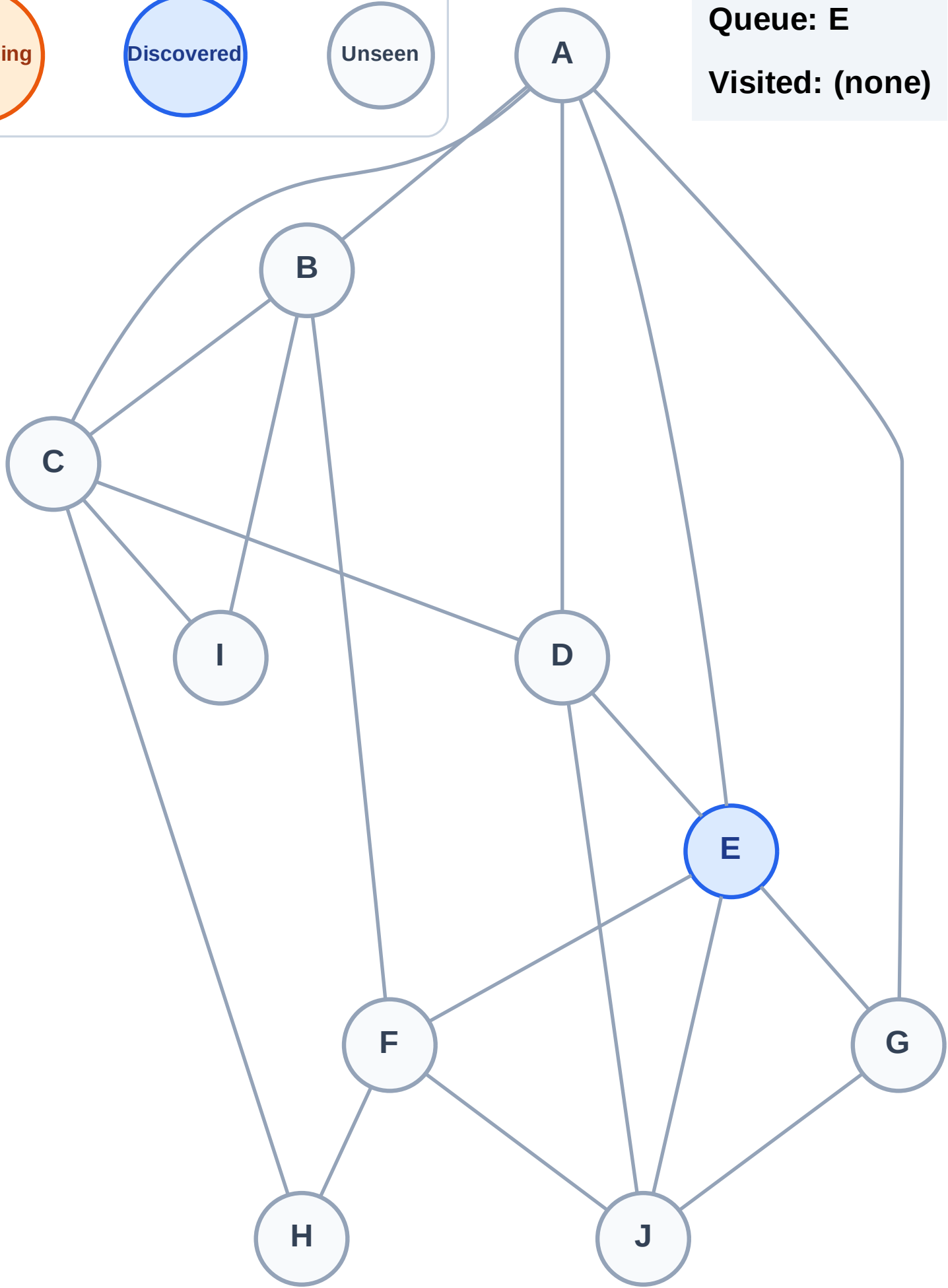
Step 1: Start at E

Legend



Queue: E

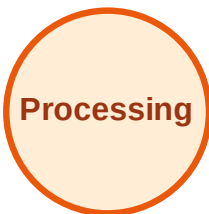
Visited: (none)



BFS Traversal

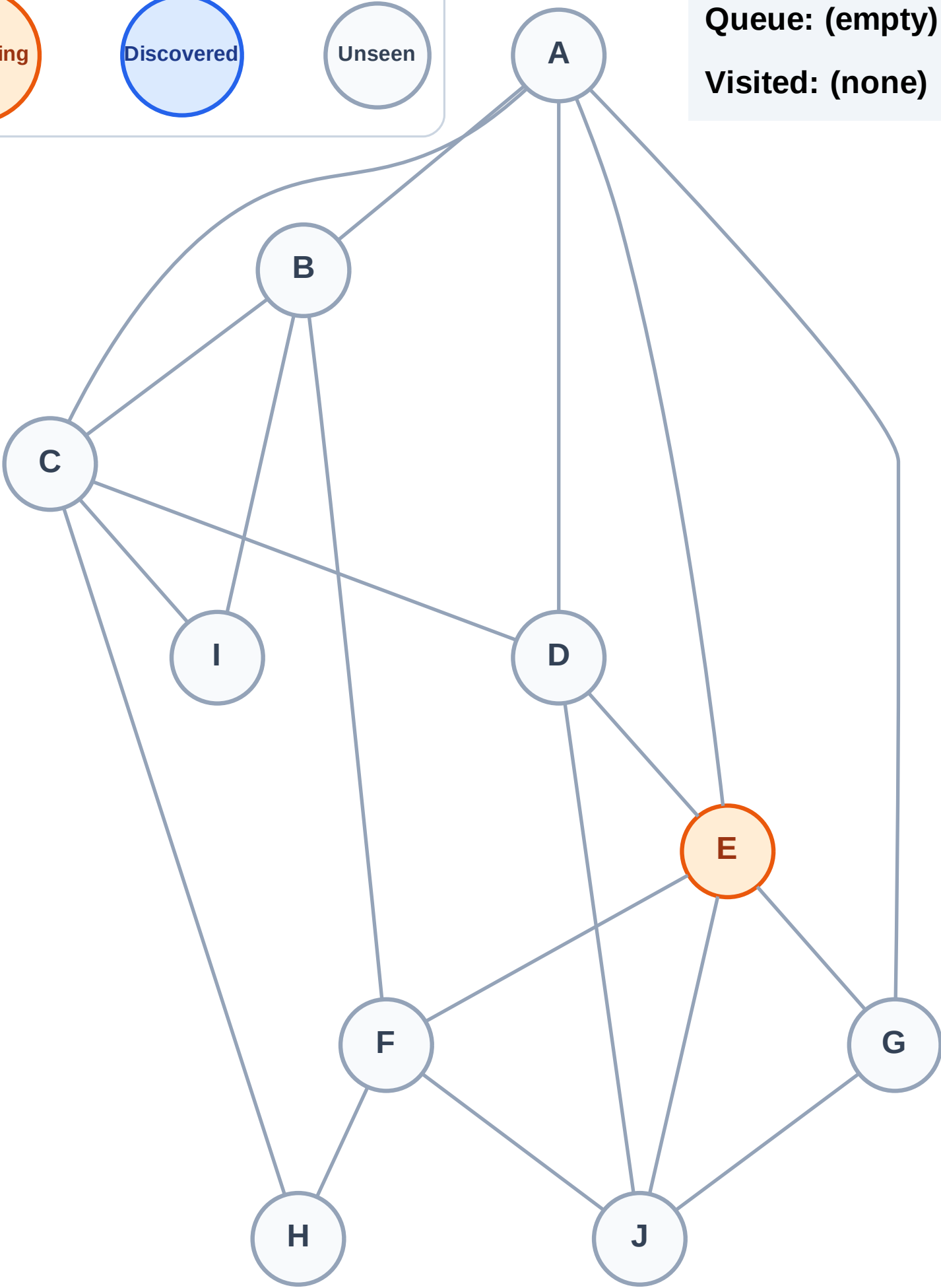
Step 2: Process node E

Legend



Queue: (empty)

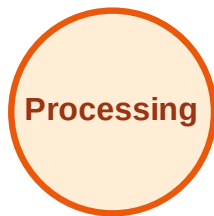
Visited: (none)



BFS Traversal

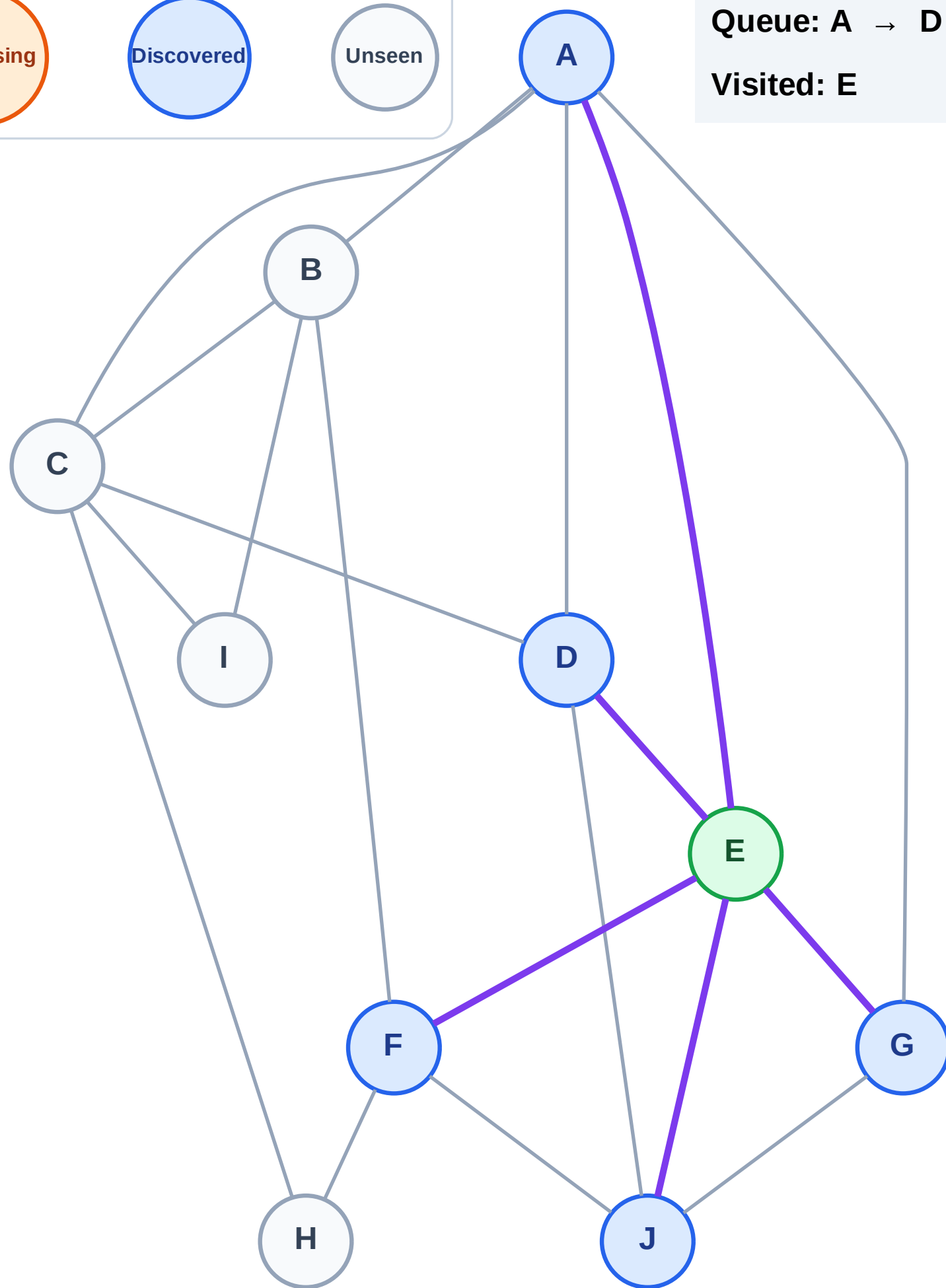
Step 3: Discover A, D, F, G, J from E

Legend



Queue: A → D → F → G → J

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

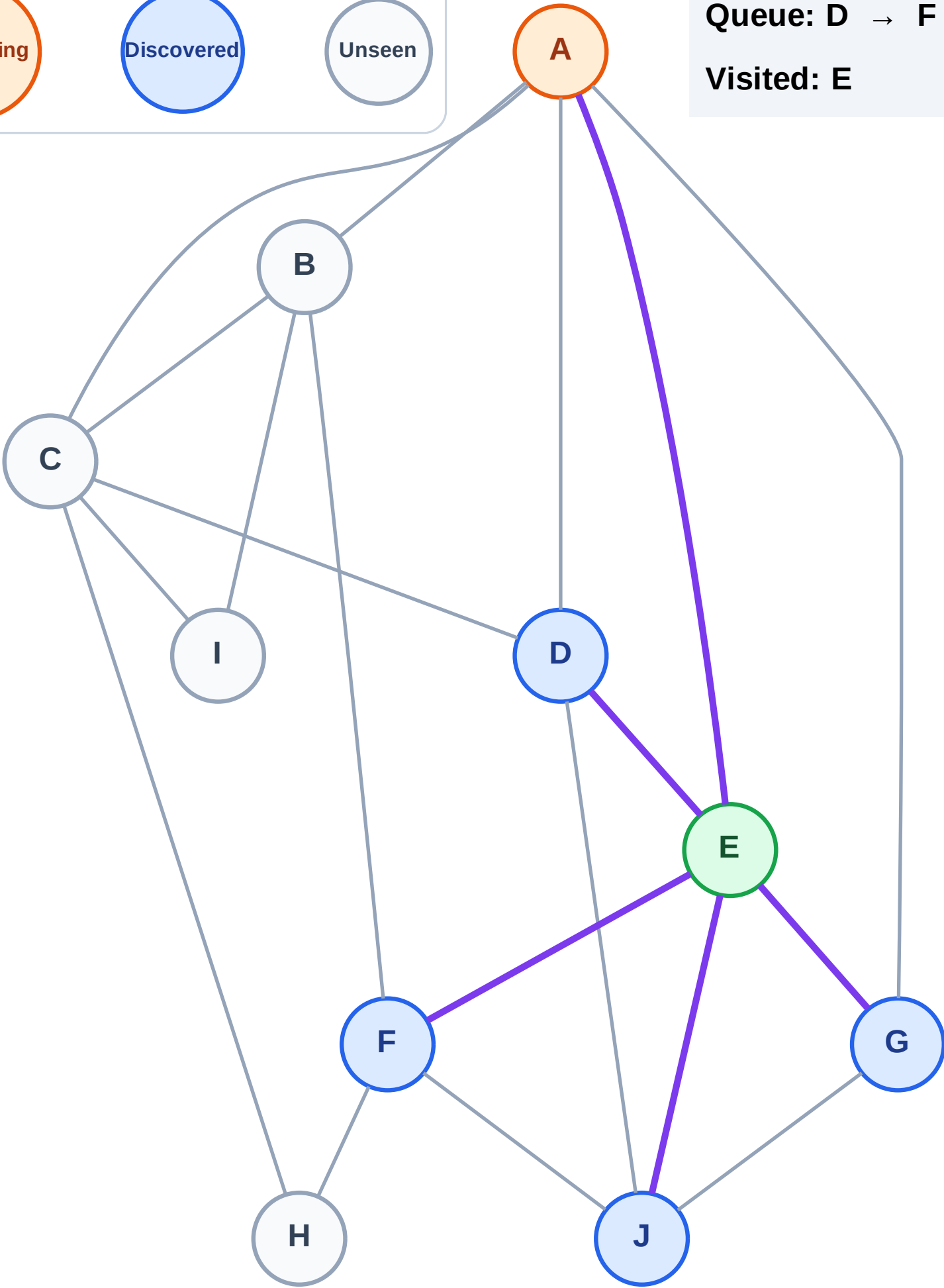
Processing

Discovered

Unseen

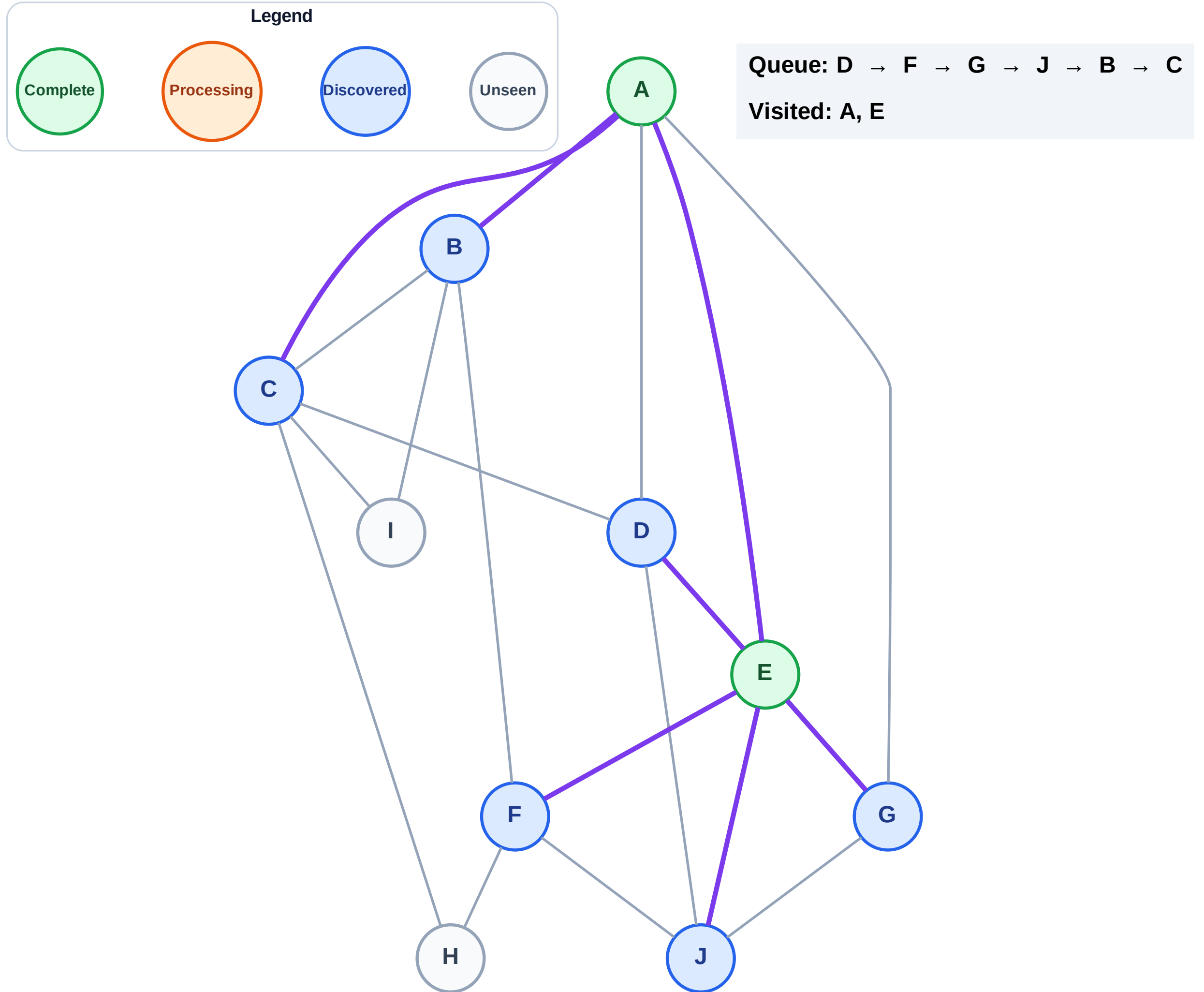
Queue: D → F → G → J

Visited: E



Step 5: Discover B, C from A

Step 5: Discover B, C from A



BFS Traversal

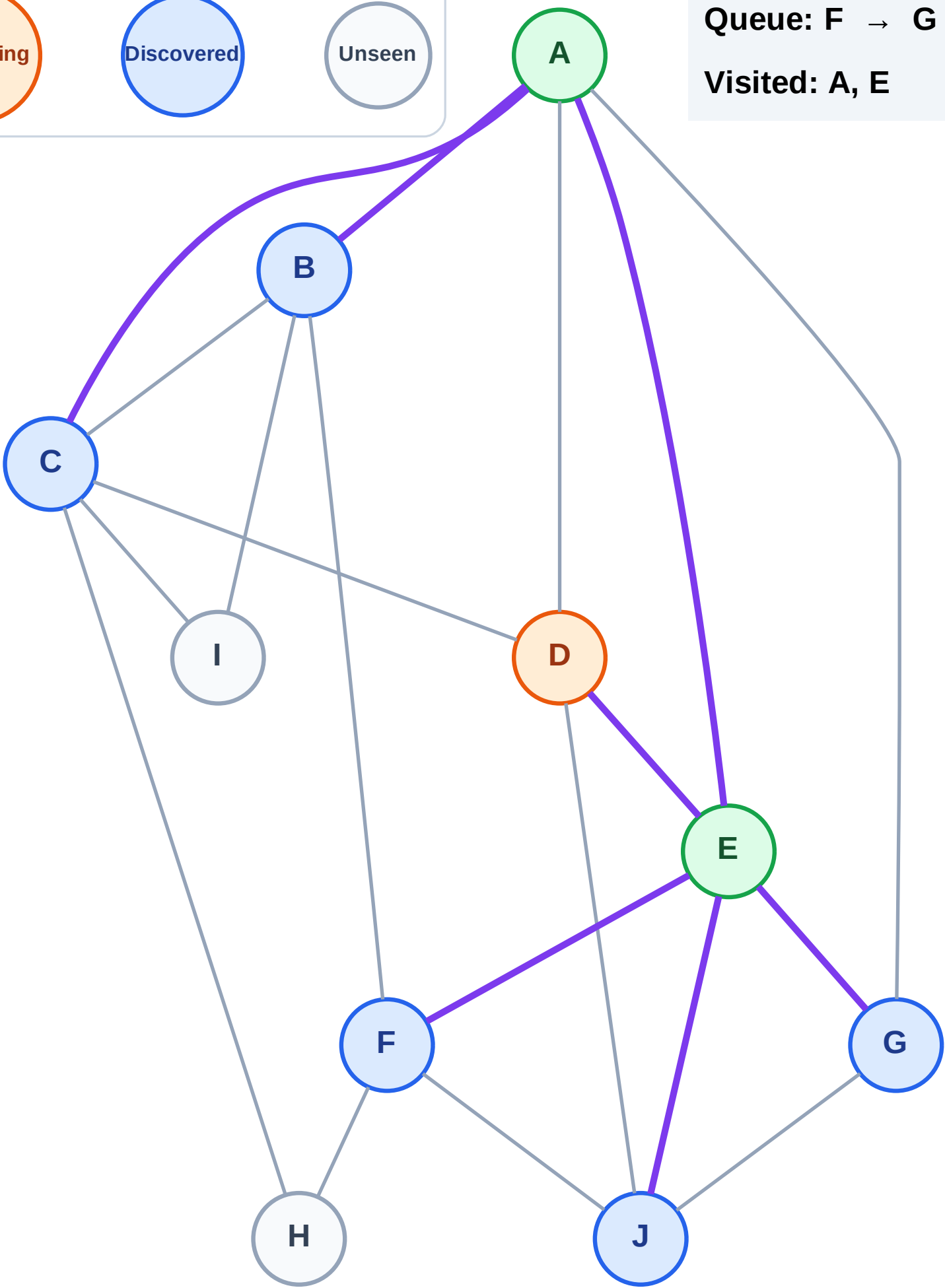
Step 6: Process node D

Legend



Queue: F → G → J → B → C

Visited: A, E



BFS Traversal

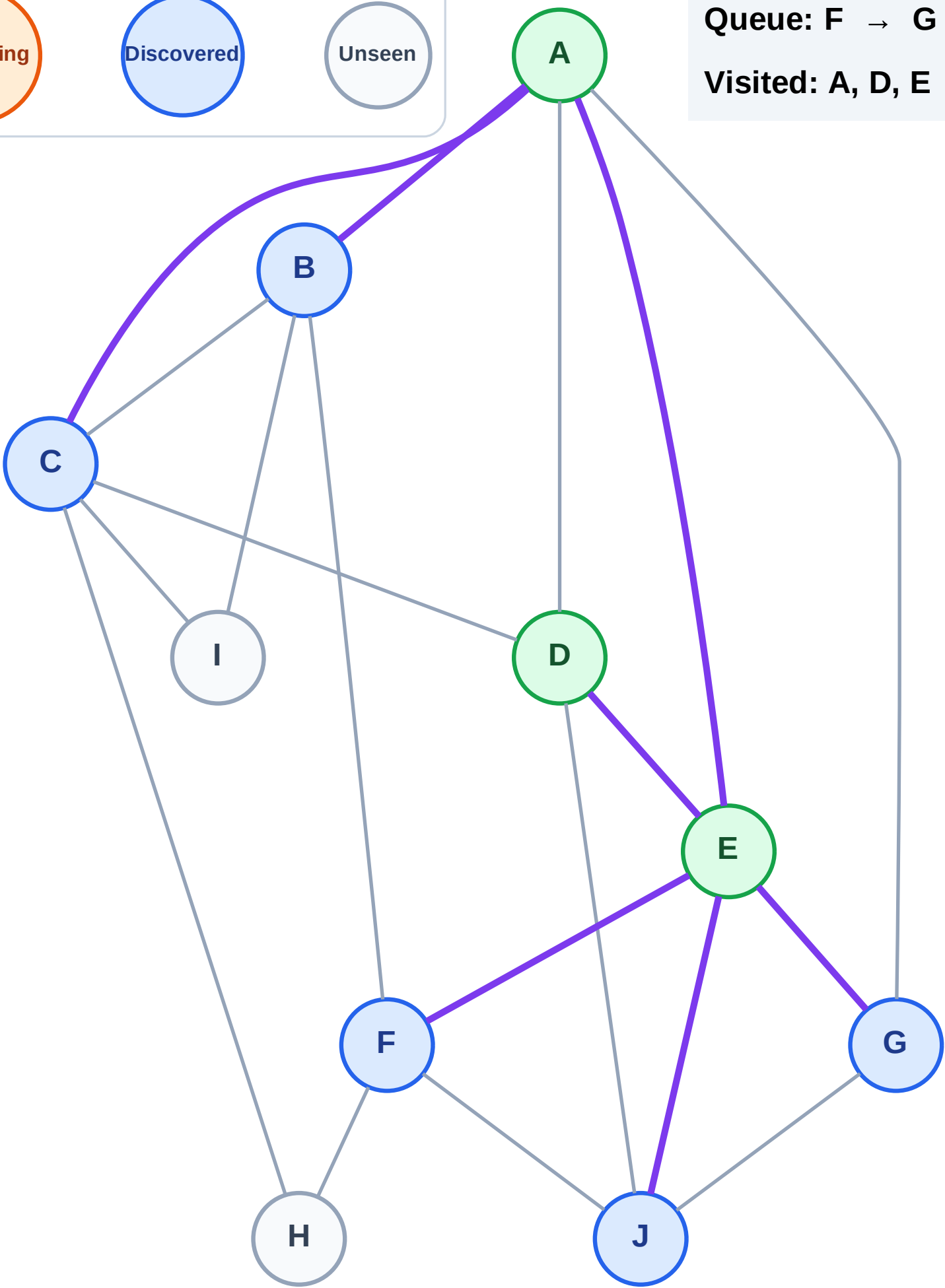
Step 7: No new neighbors from D

Legend



Queue: F → G → J → B → C

Visited: A, D, E



BFS Traversal

Step 8: Process node F

Legend

Complete

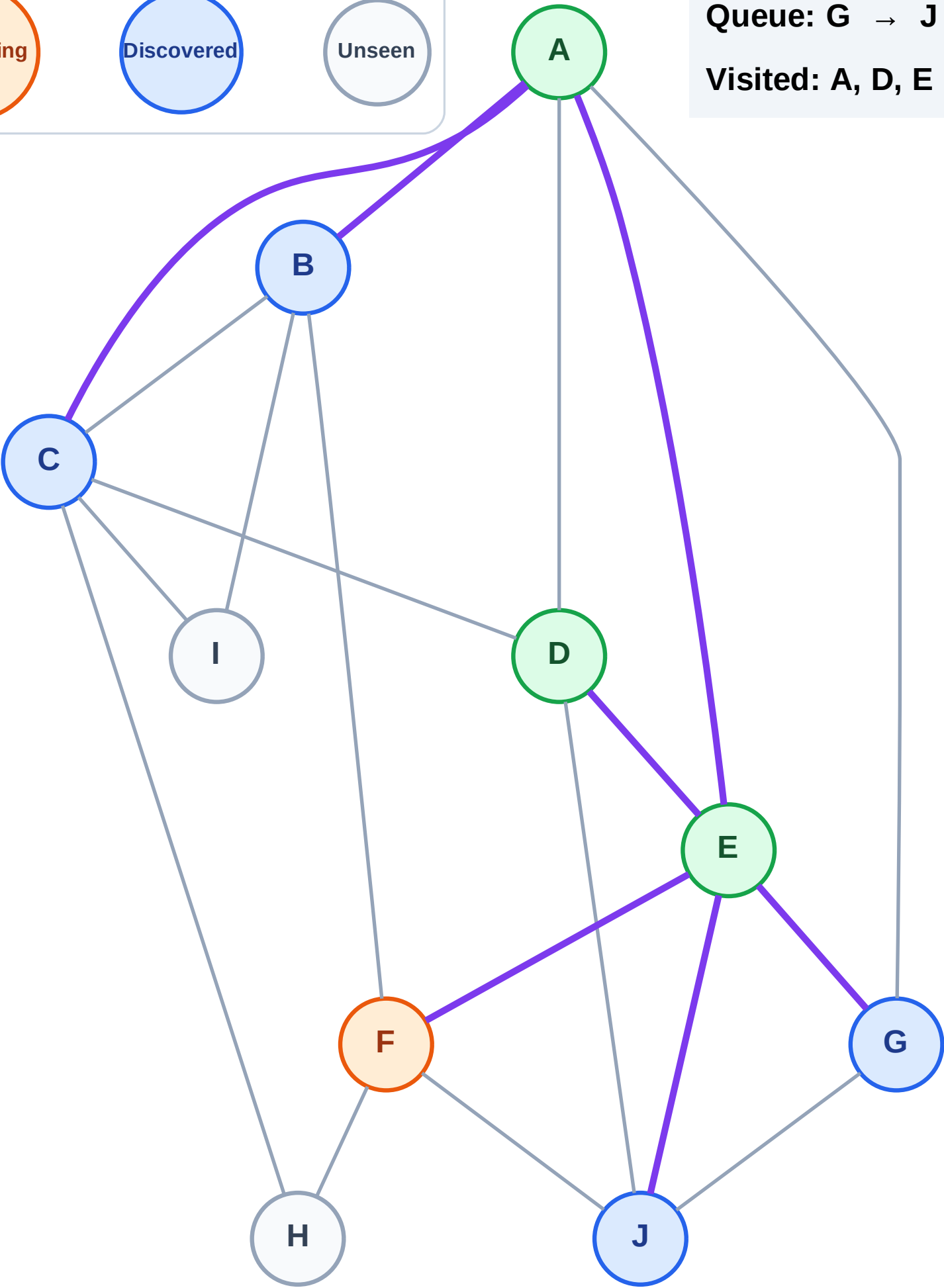
Processing

Discovered

Unseen

Queue: G → J → B → C

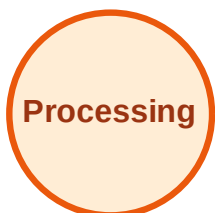
Visited: A, D, E



BFS Traversal

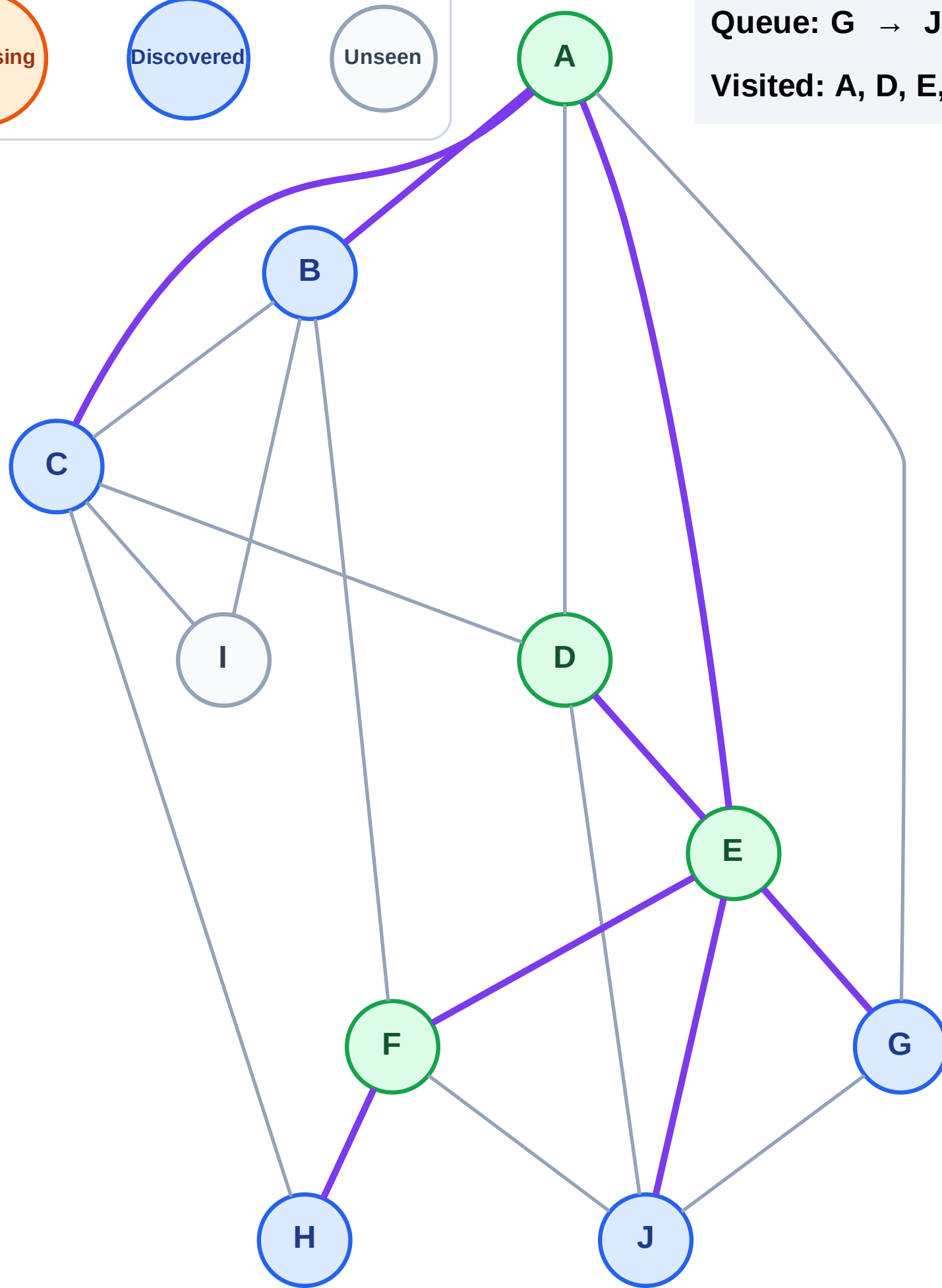
Step 9: Discover H from F

Legend



Queue: G → J → B → C → H

Visited: A, D, E, F



BFS Traversal

Step 10: Process node G

Legend

Complete

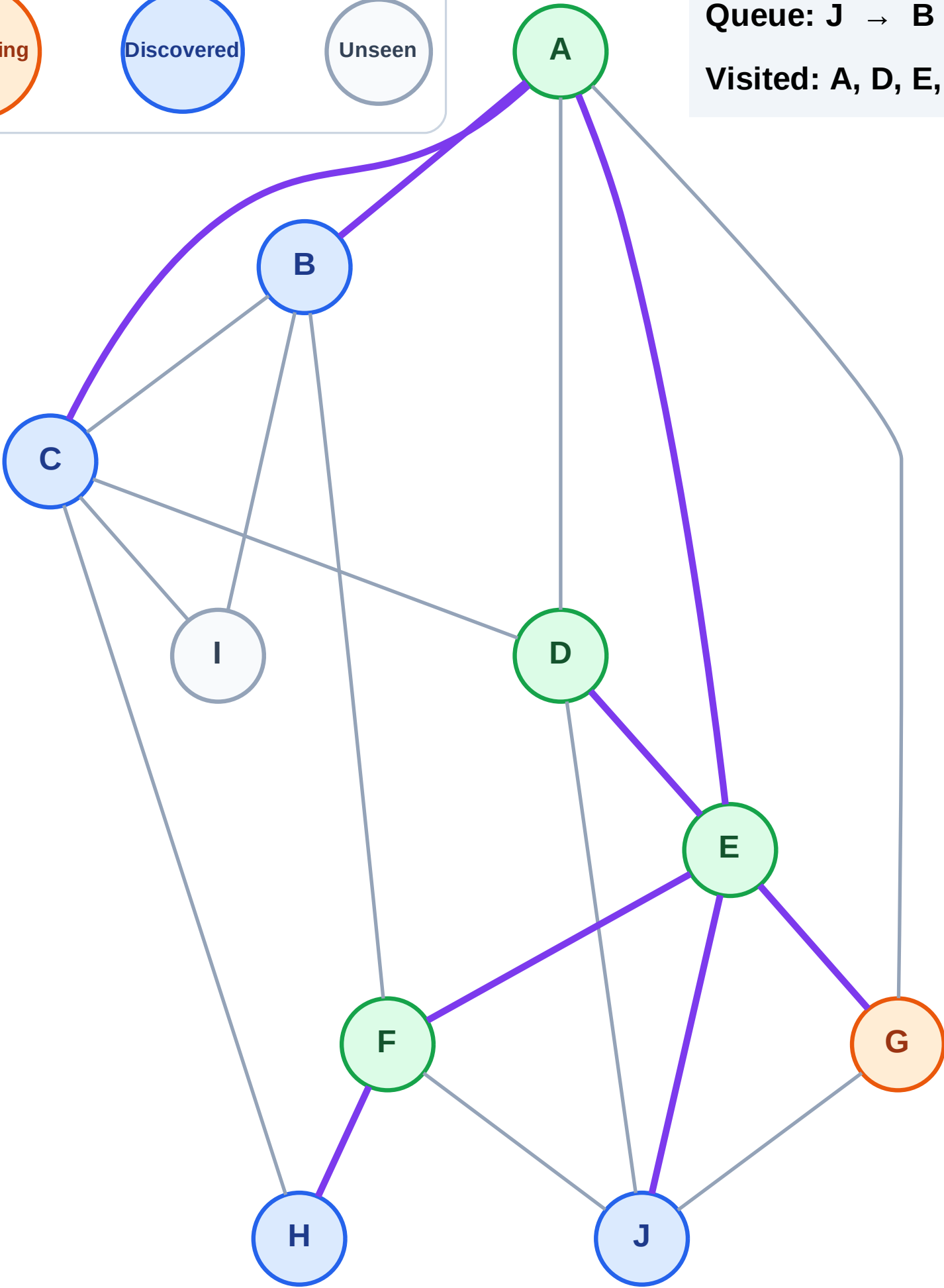
Processing

Discovered

Unseen

Queue: J → B → C → H

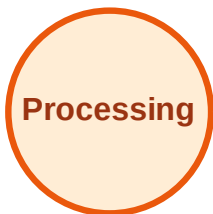
Visited: A, D, E, F



BFS Traversal

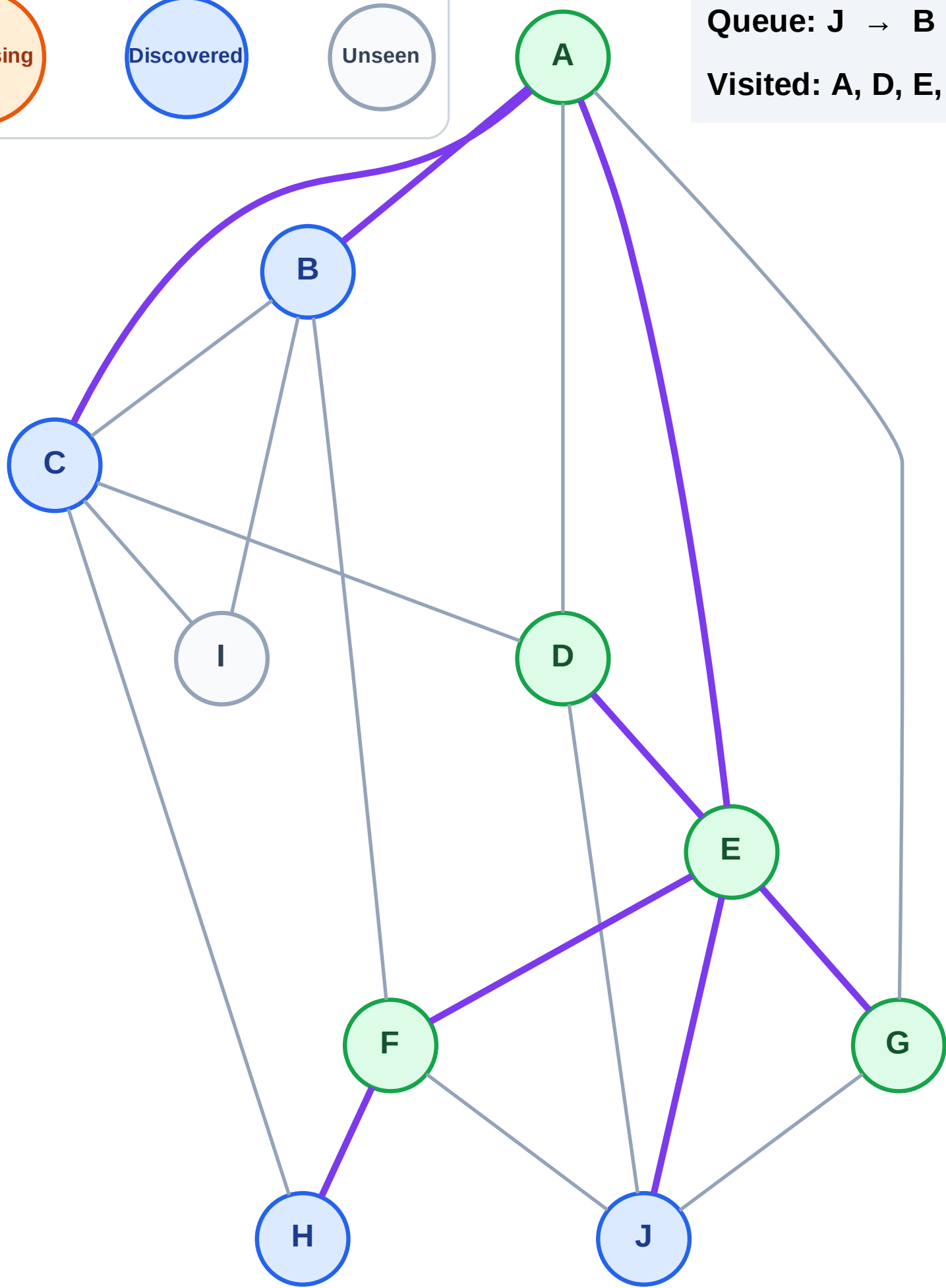
Step 11: No new neighbors from G

Legend



Queue: J → B → C → H

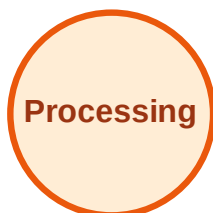
Visited: A, D, E, F, G



BFS Traversal

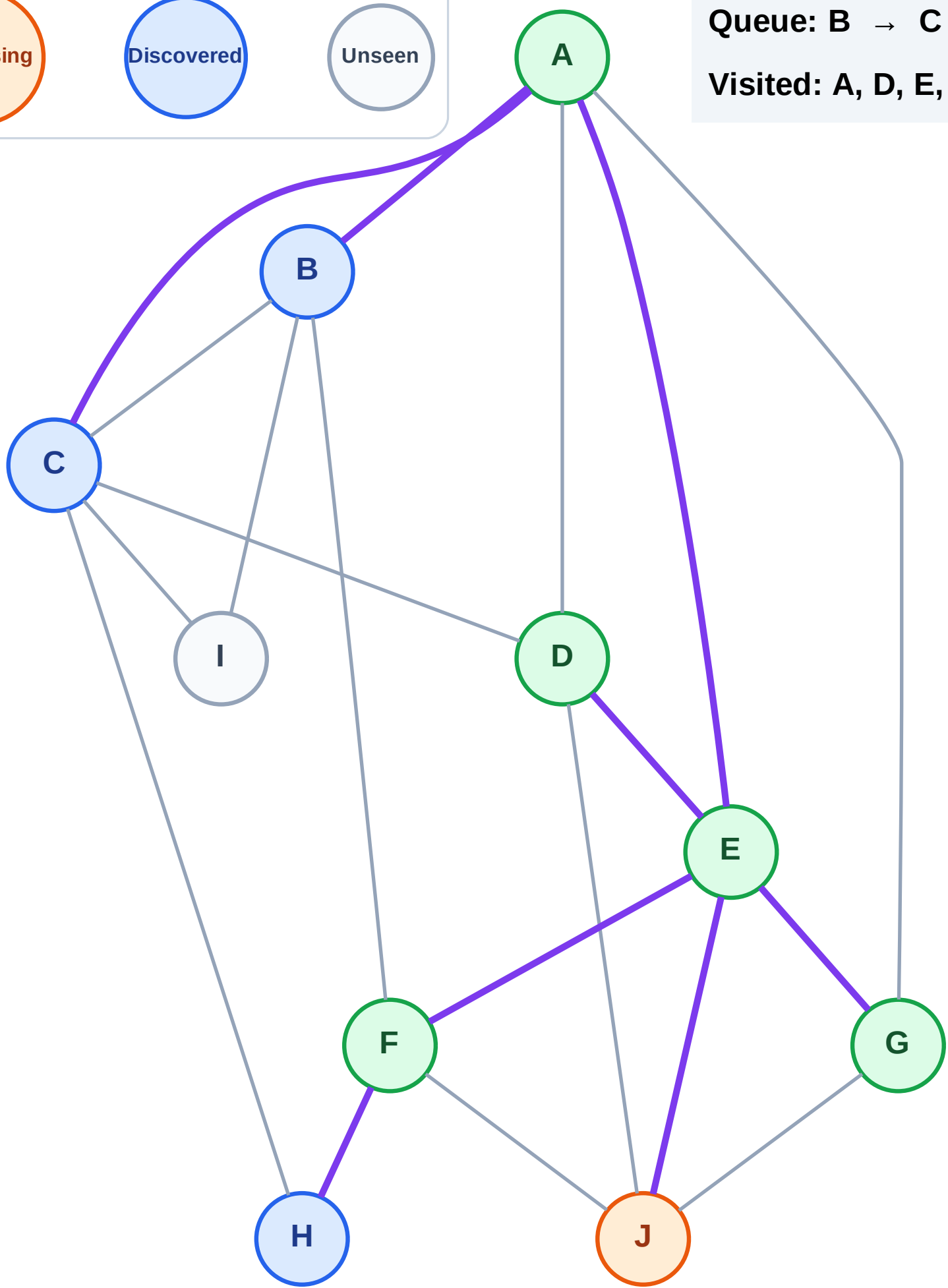
Step 12: Process node J

Legend



Queue: B → C → H

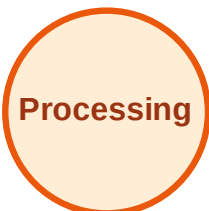
Visited: A, D, E, F, G



BFS Traversal

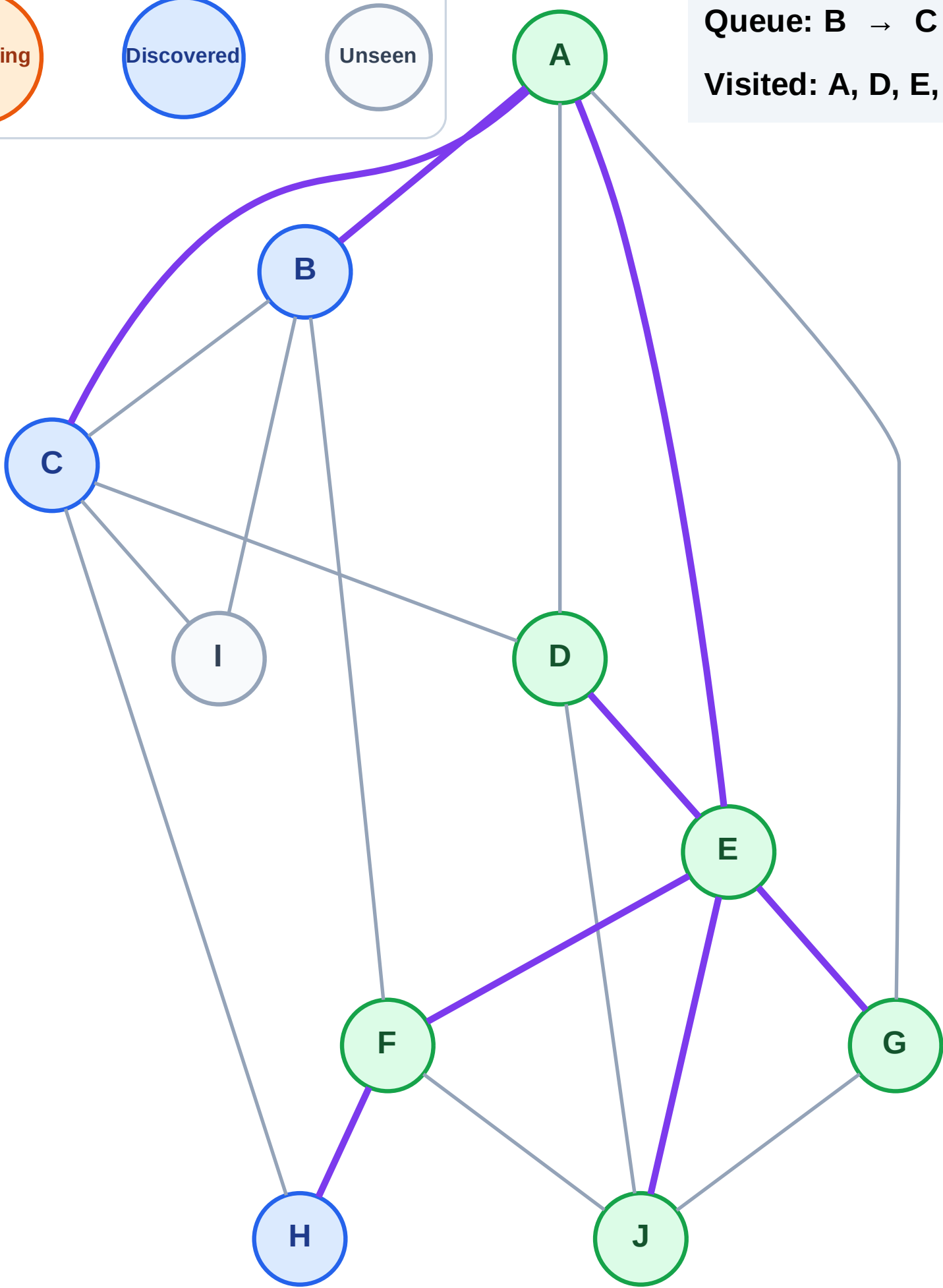
Step 13: No new neighbors from J

Legend



Queue: B → C → H

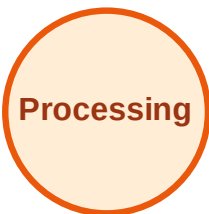
Visited: A, D, E, F, G, J



BFS Traversal

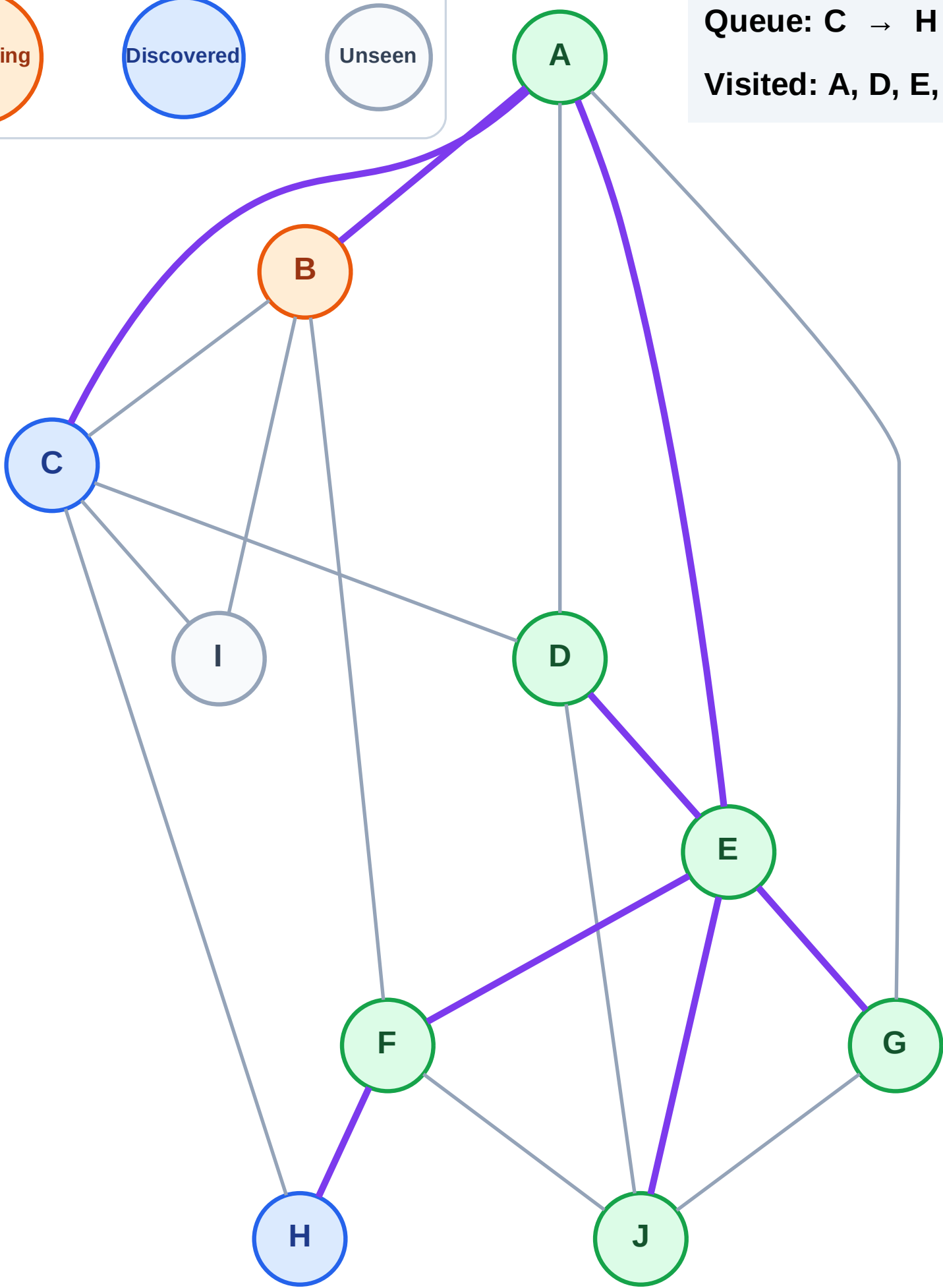
Step 14: Process node B

Legend



Queue: C → H

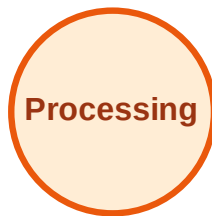
Visited: A, D, E, F, G, J



BFS Traversal

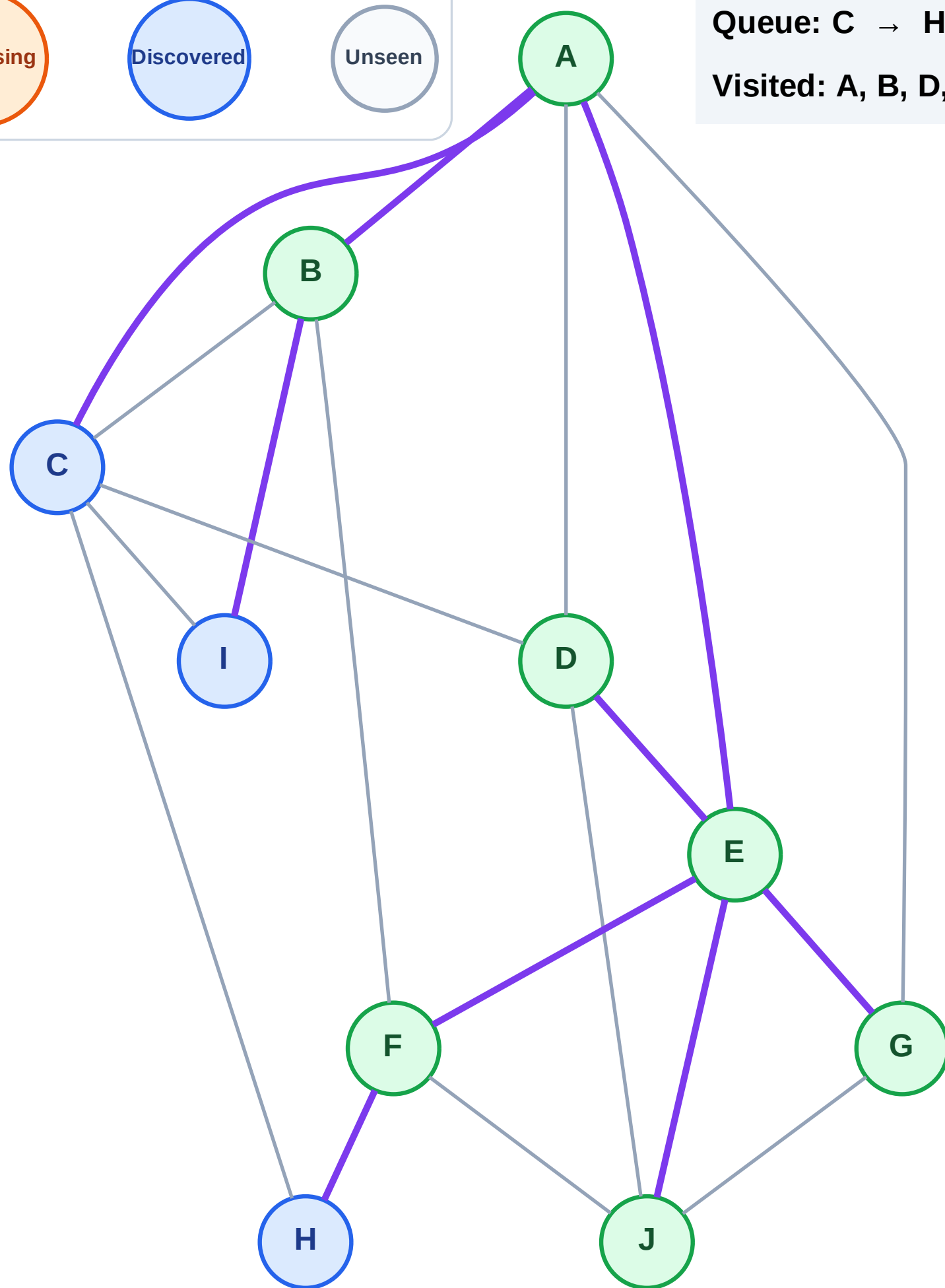
Step 15: Discover I from B

Legend



Queue: C → H → I

Visited: A, B, D, E, F, G, J



BFS Traversal

Step 16: Process node C

Legend

Complete

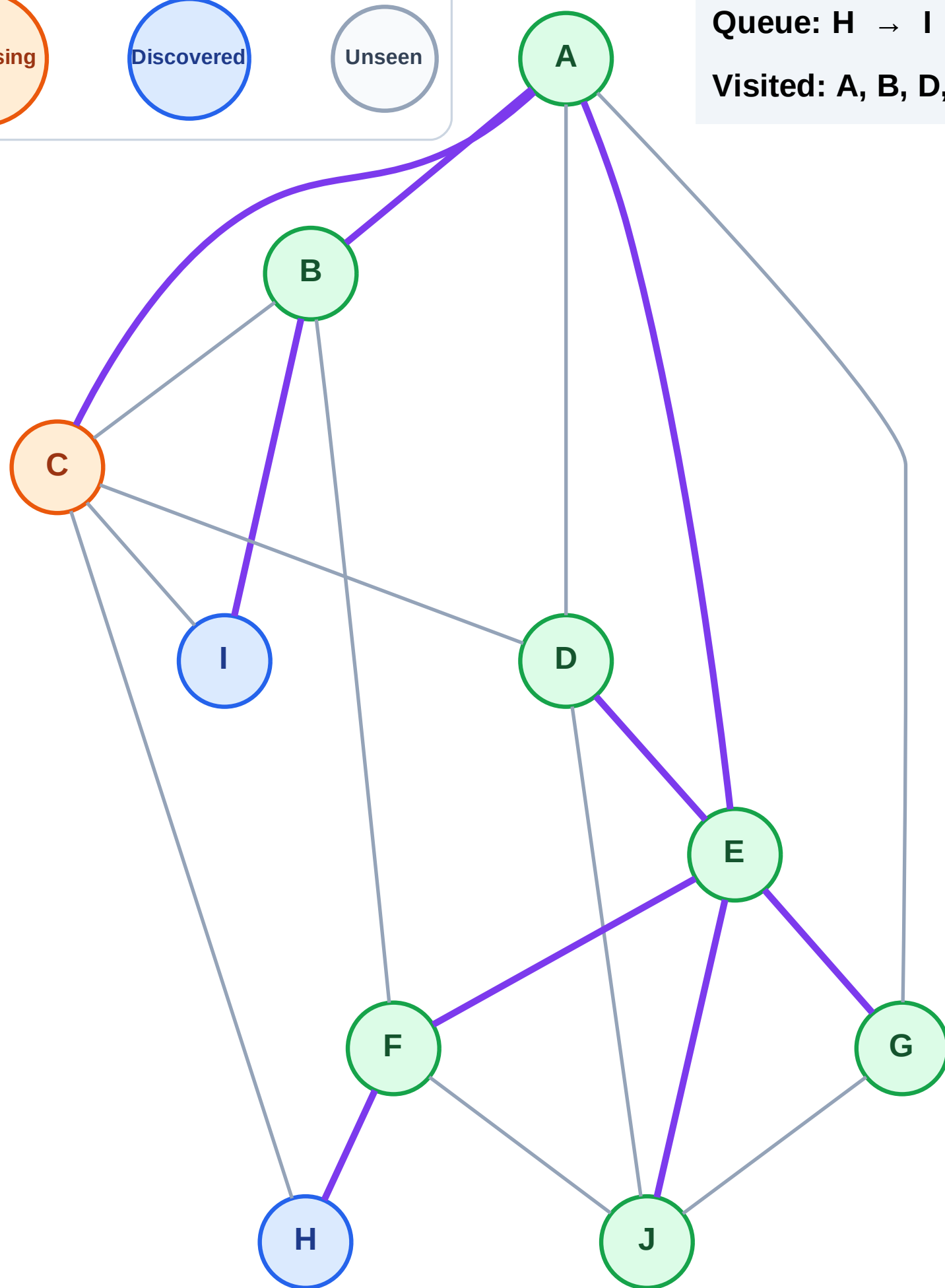
Processing

Discovered

Unseen

Queue: H → I

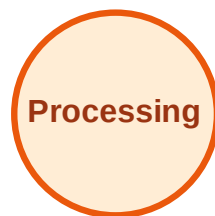
Visited: A, B, D, E, F, G, J



BFS Traversal

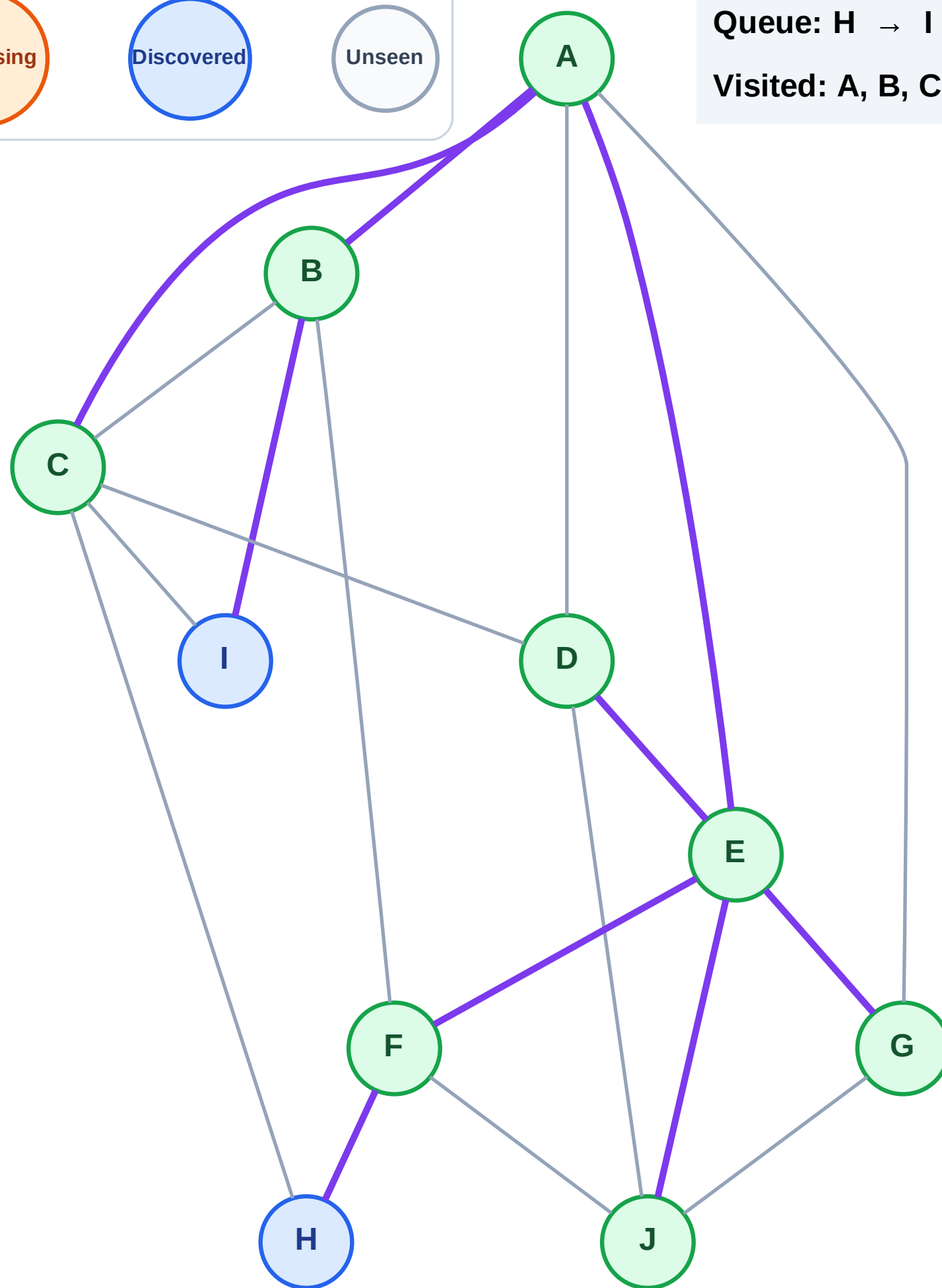
Step 17: No new neighbors from C

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node H

Legend

Complete

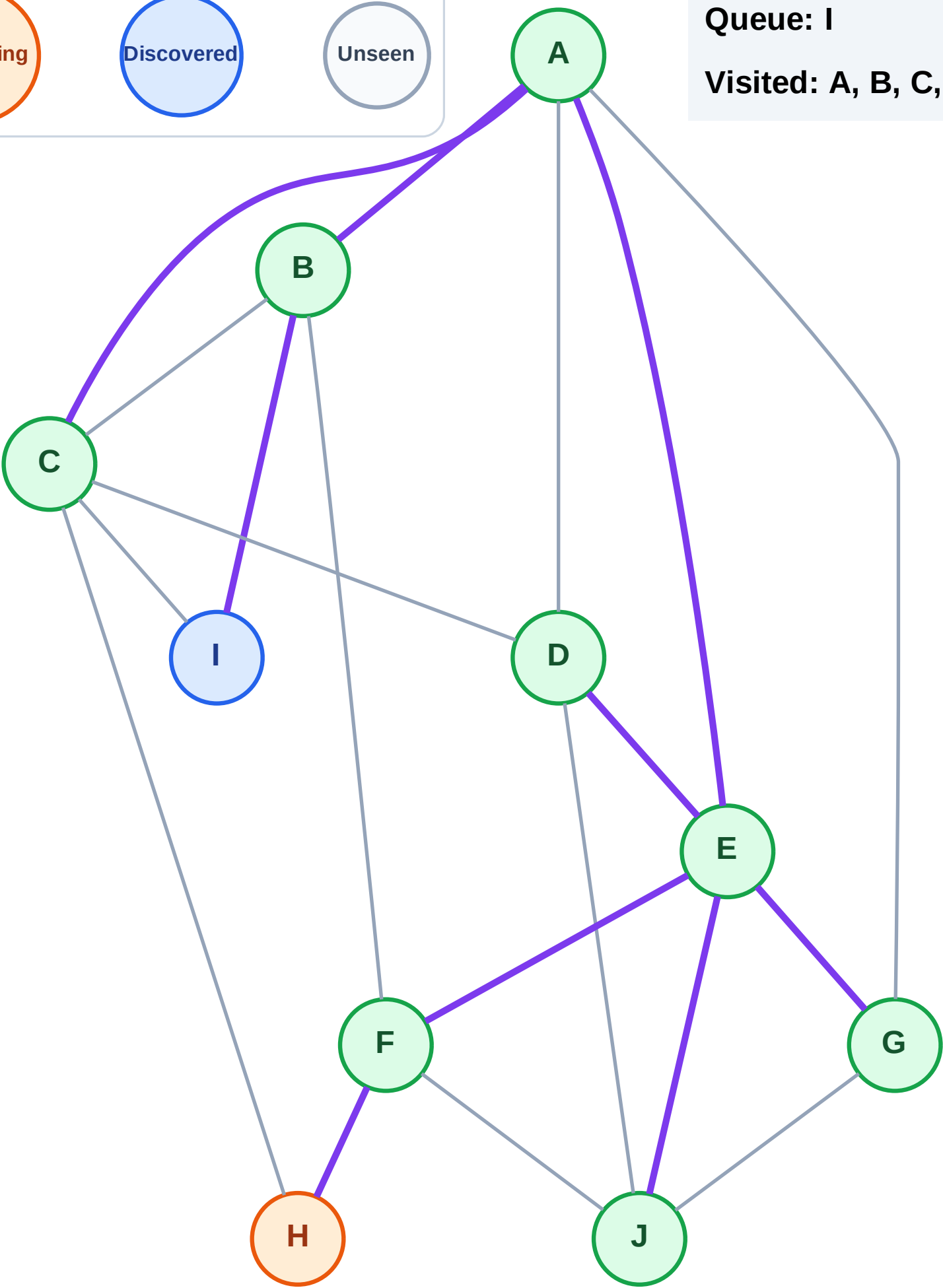
Processing

Discovered

Unseen

Queue: I

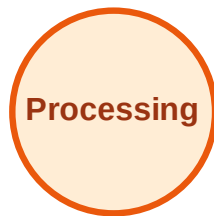
Visited: A, B, C, D, E, F, G, J



BFS Traversal

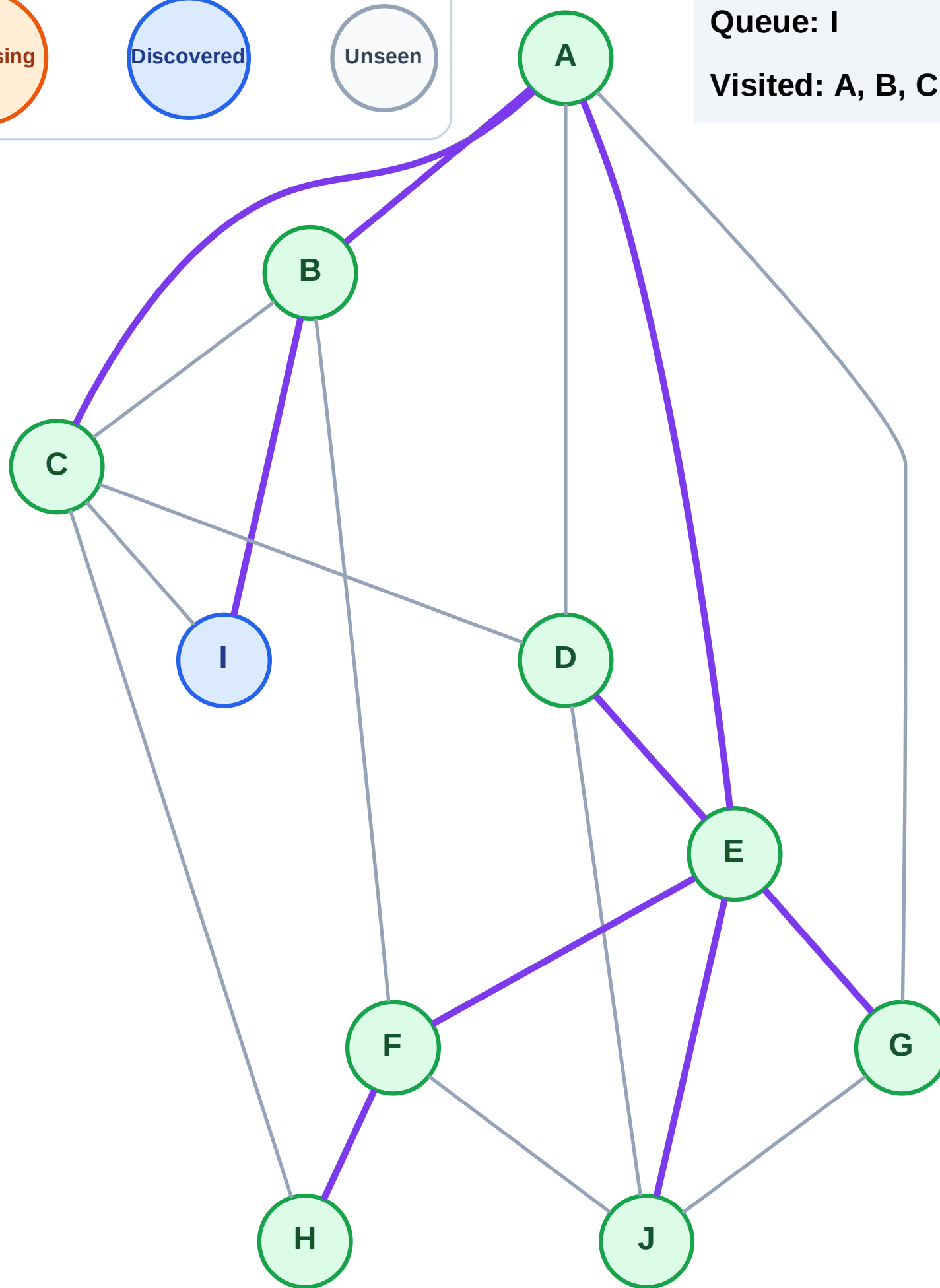
Step 19: No new neighbors from H

Legend



Queue: I

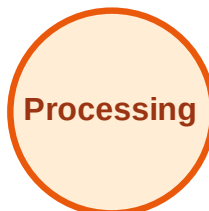
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

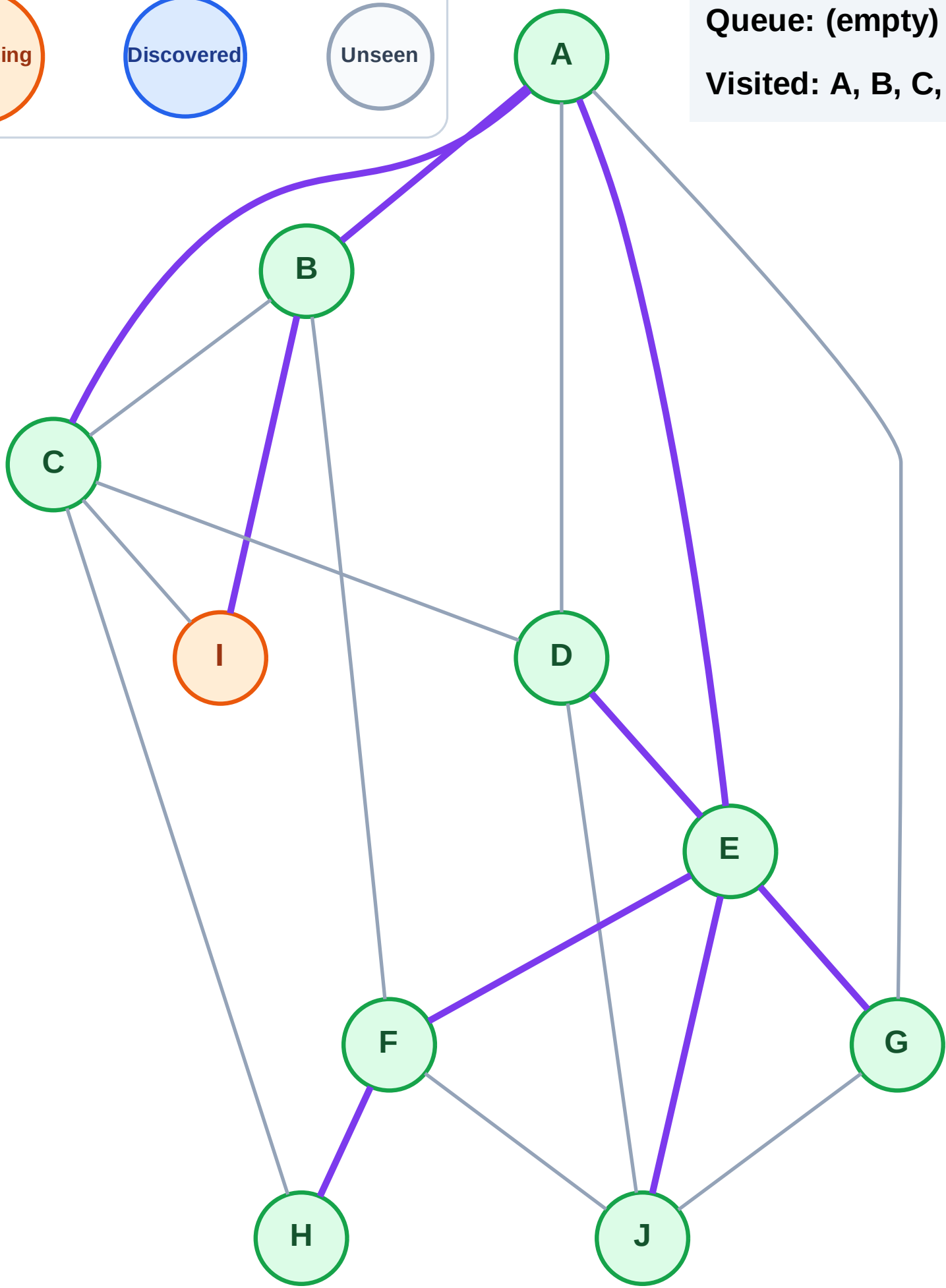
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

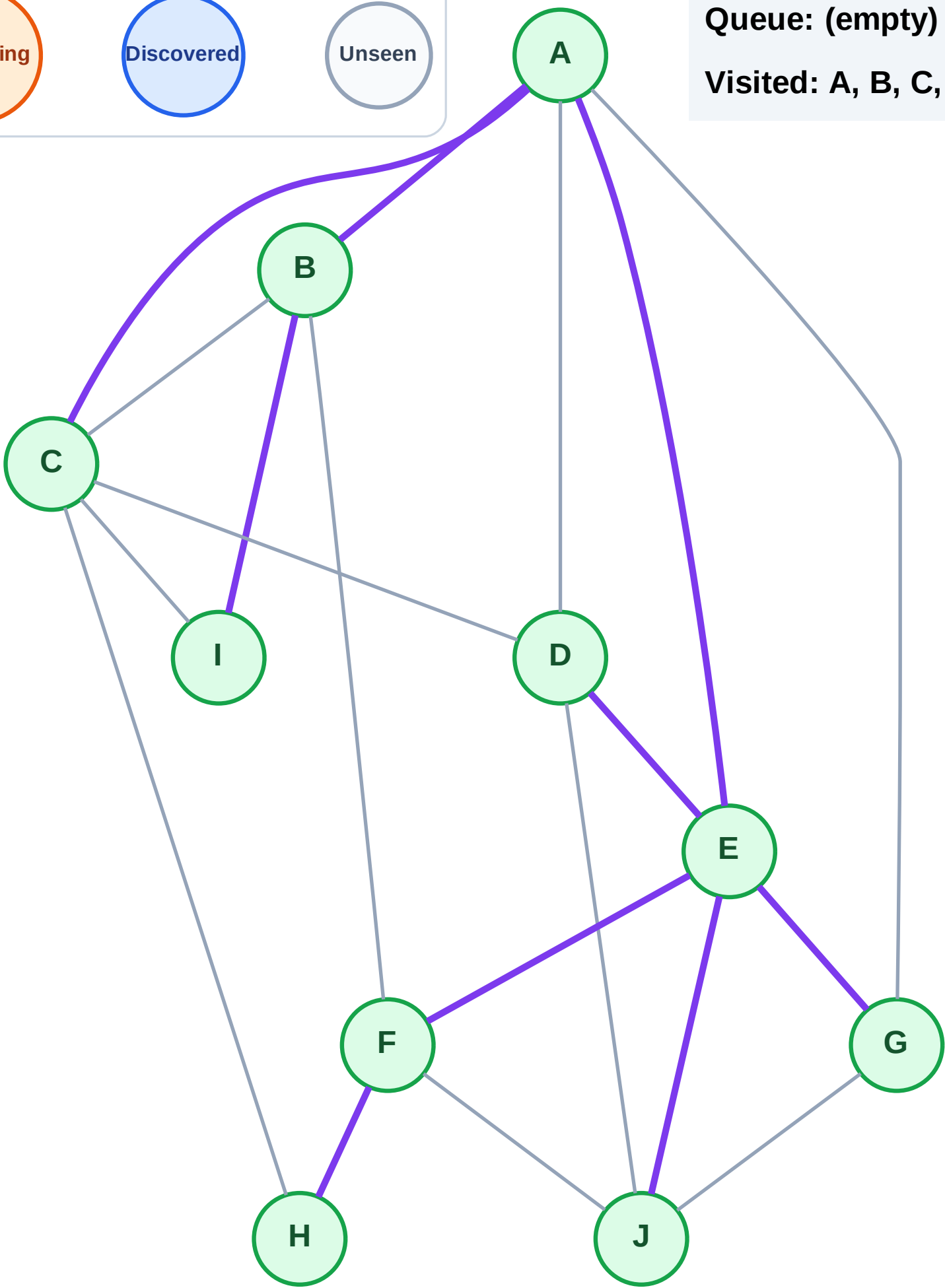
Step 21: No new neighbors from I

Legend



Queue: (empty)

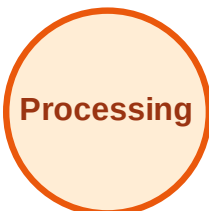
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

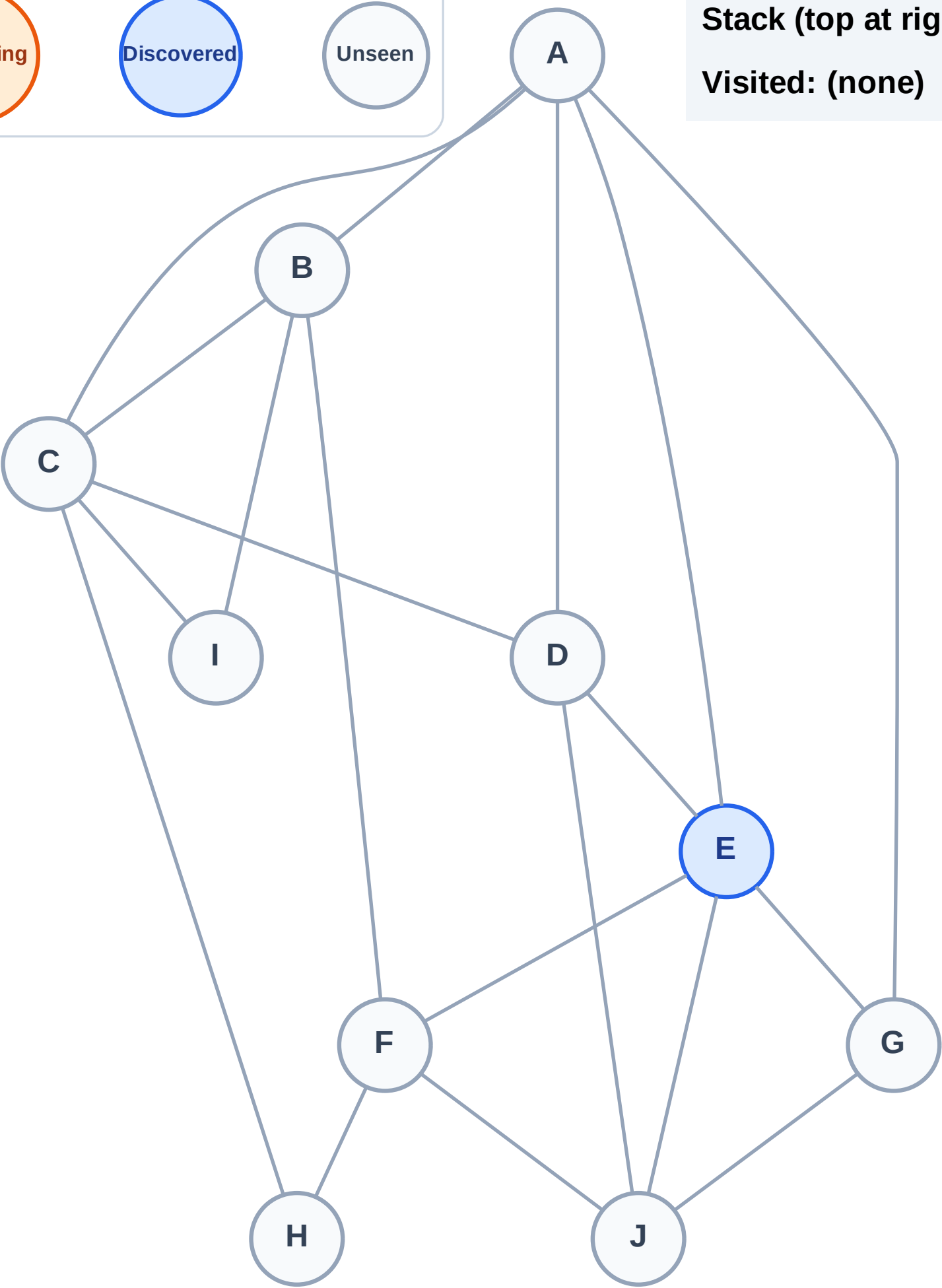
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

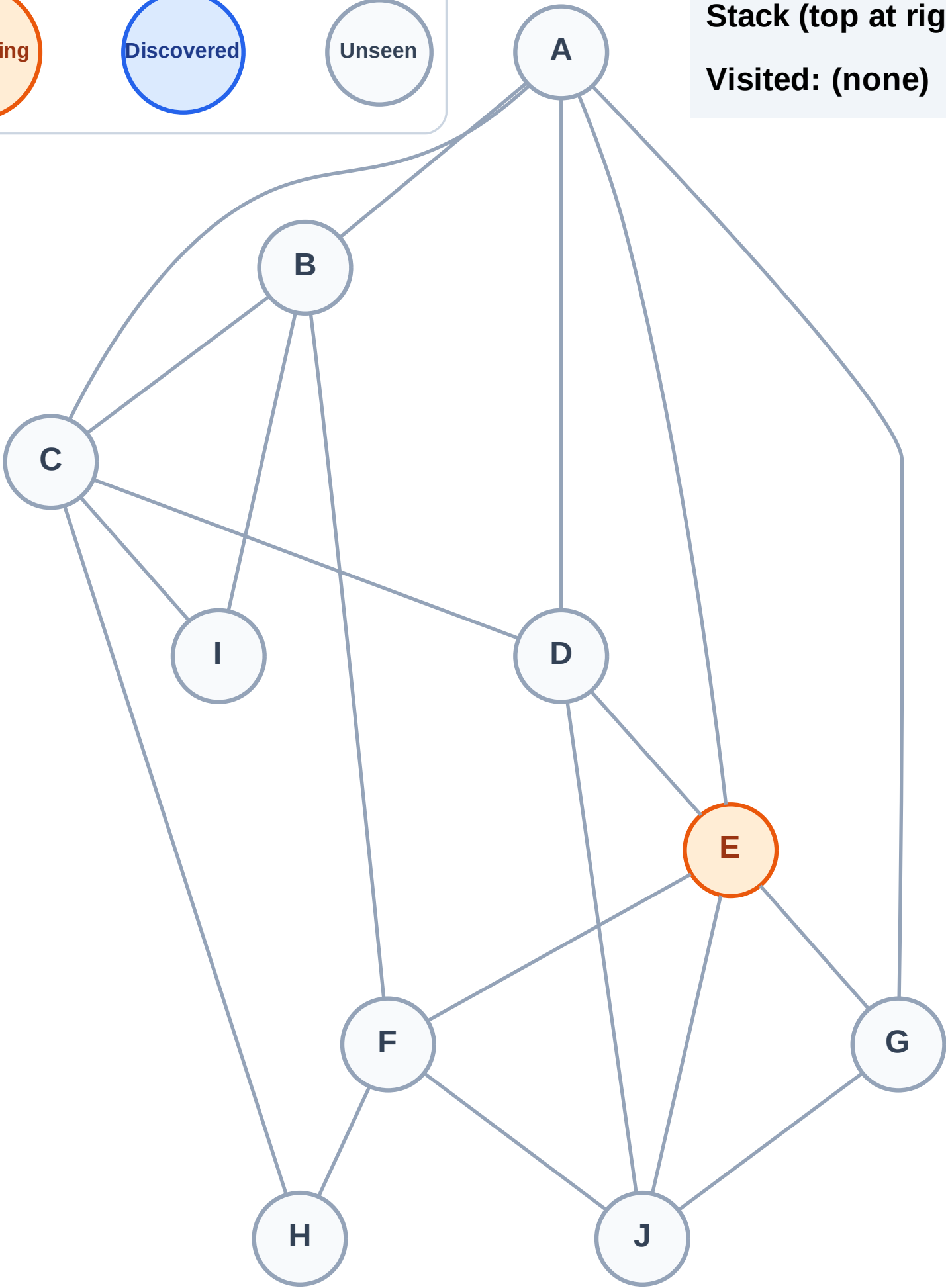
Complete

Processing

Discovered

Unseen

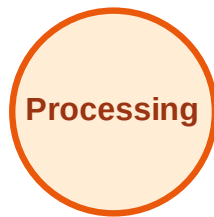
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

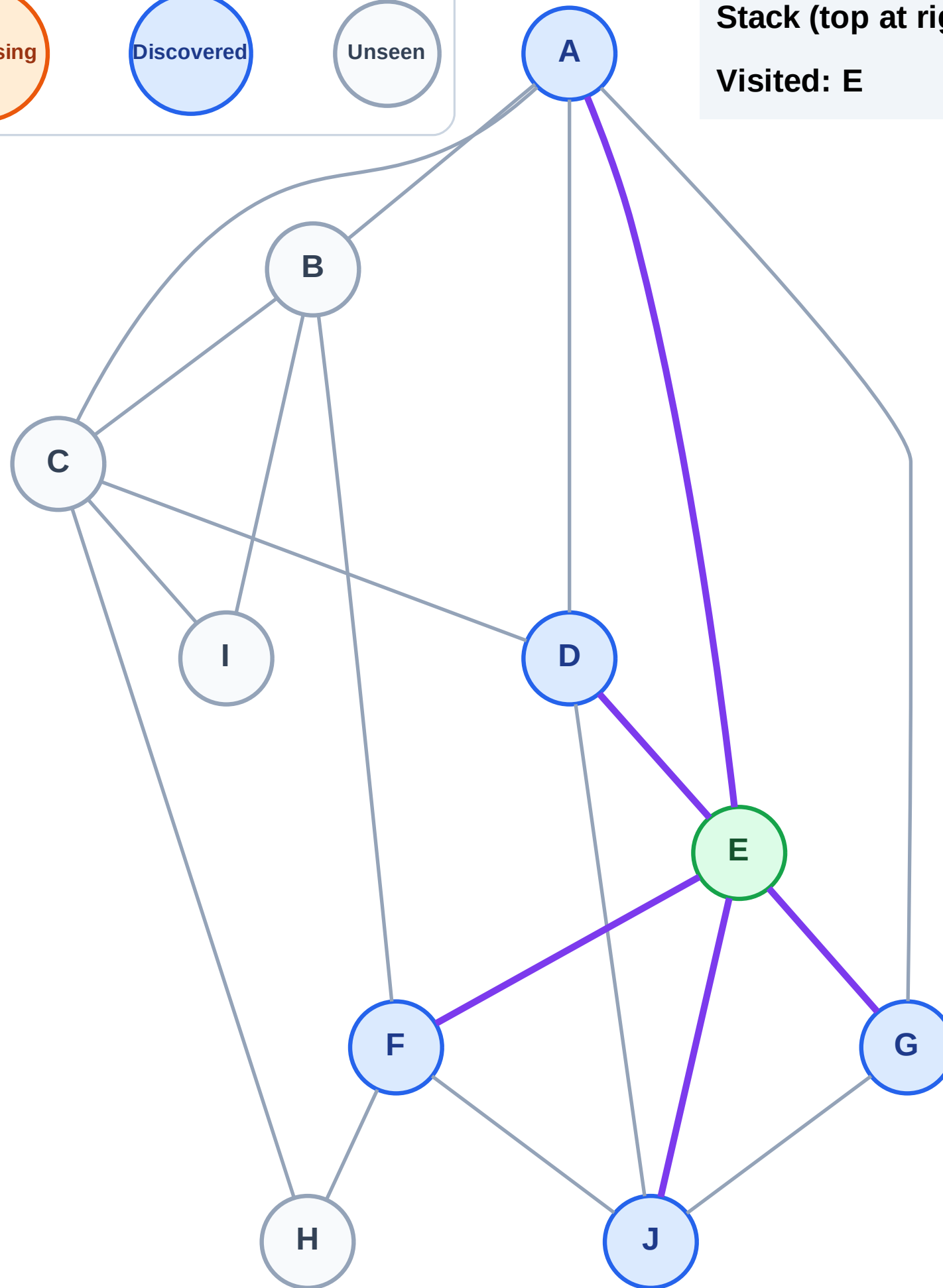
Step 3: Discover J, G, F, D, A from E

Legend



Stack (top at right): J | G | F | D | A

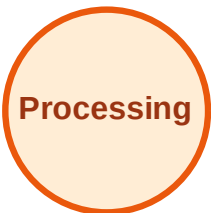
Visited: E



DFS Traversal

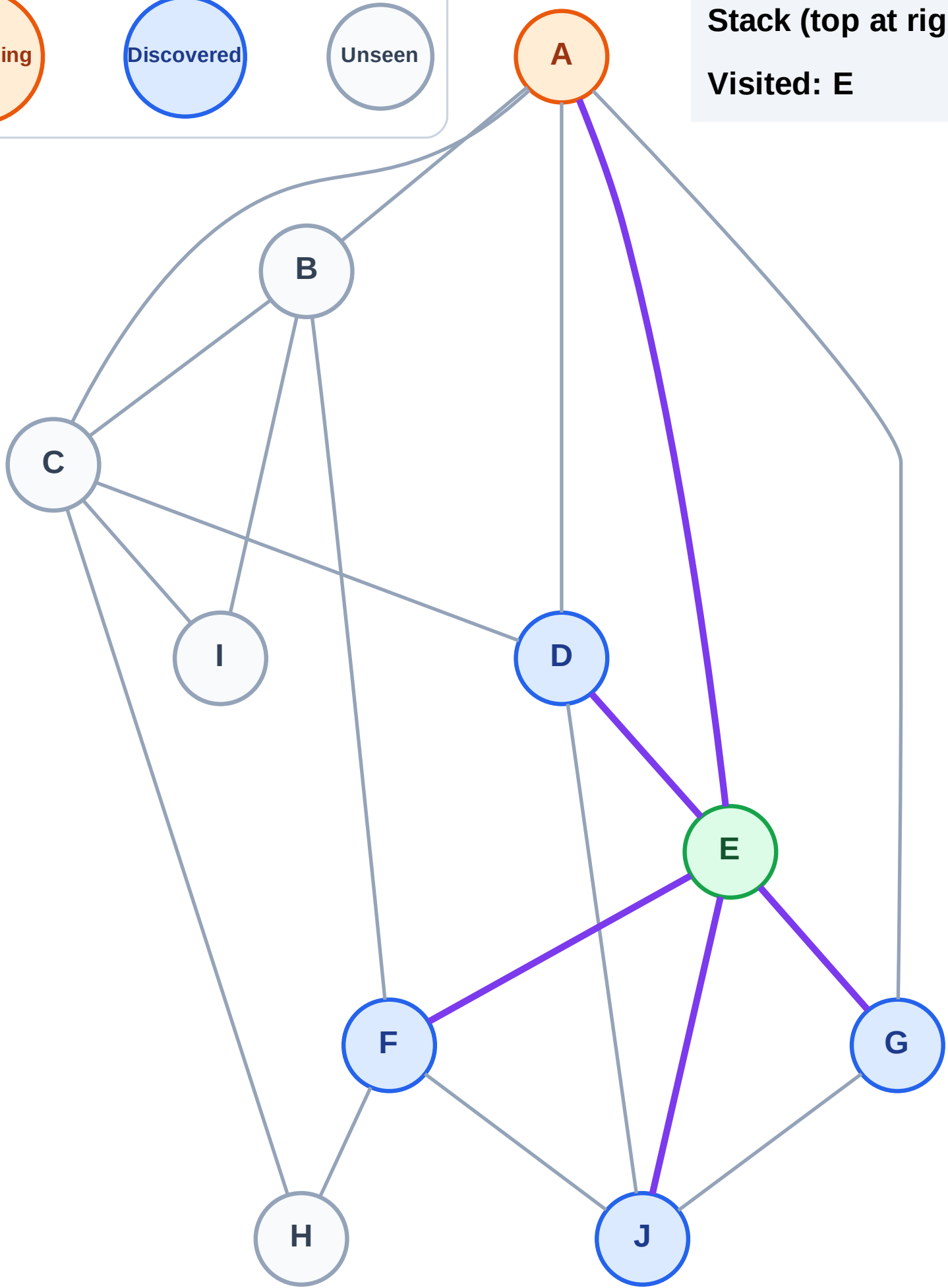
Step 4: Process node A

Legend



Stack (top at right): J | G | F | D

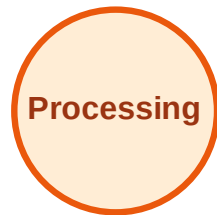
Visited: E



DFS Traversal

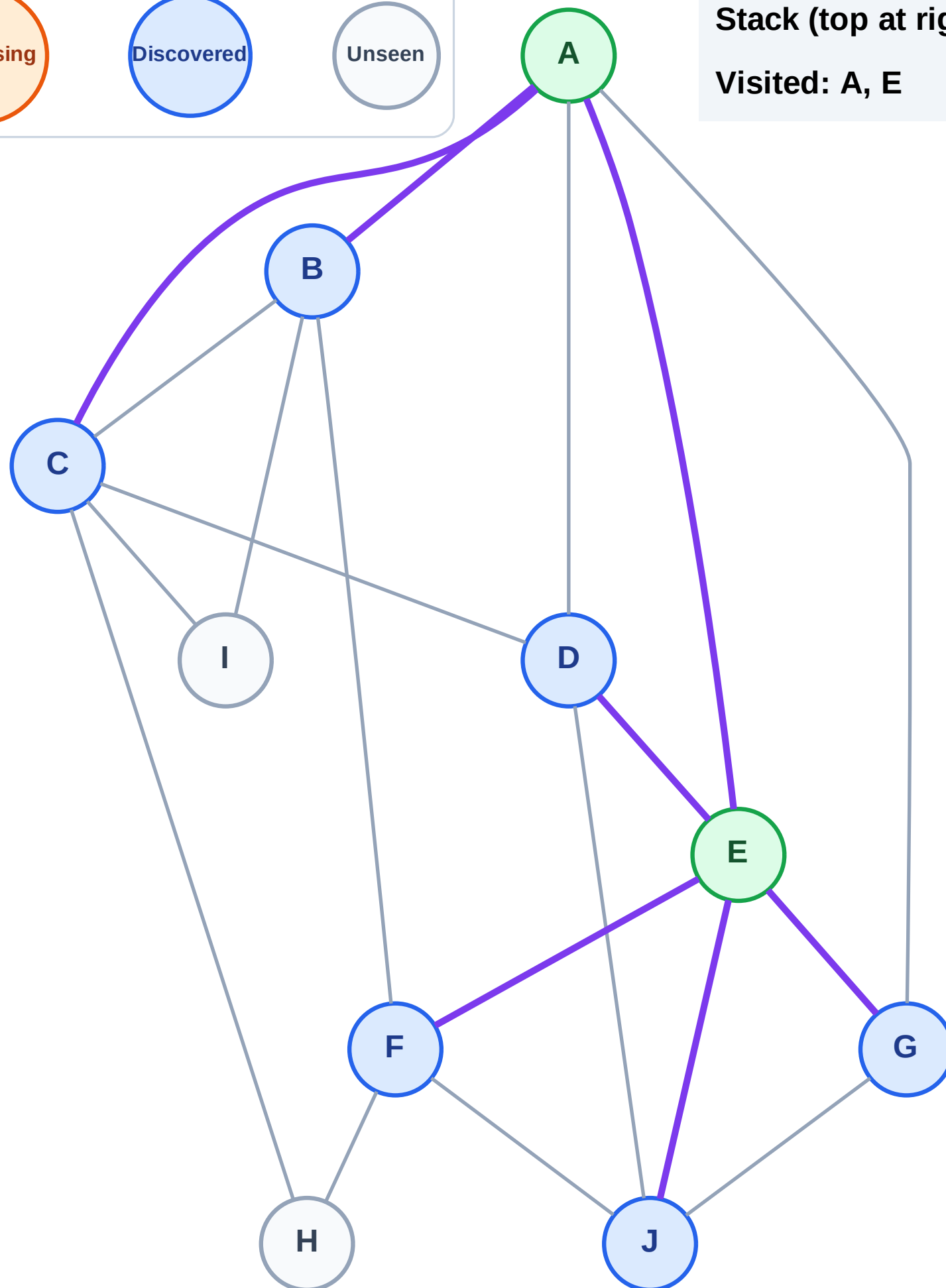
Step 5: Discover C, B from A

Legend



Stack (top at right): J | G | F | D | C | B

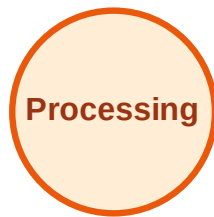
Visited: A, E



DFS Traversal

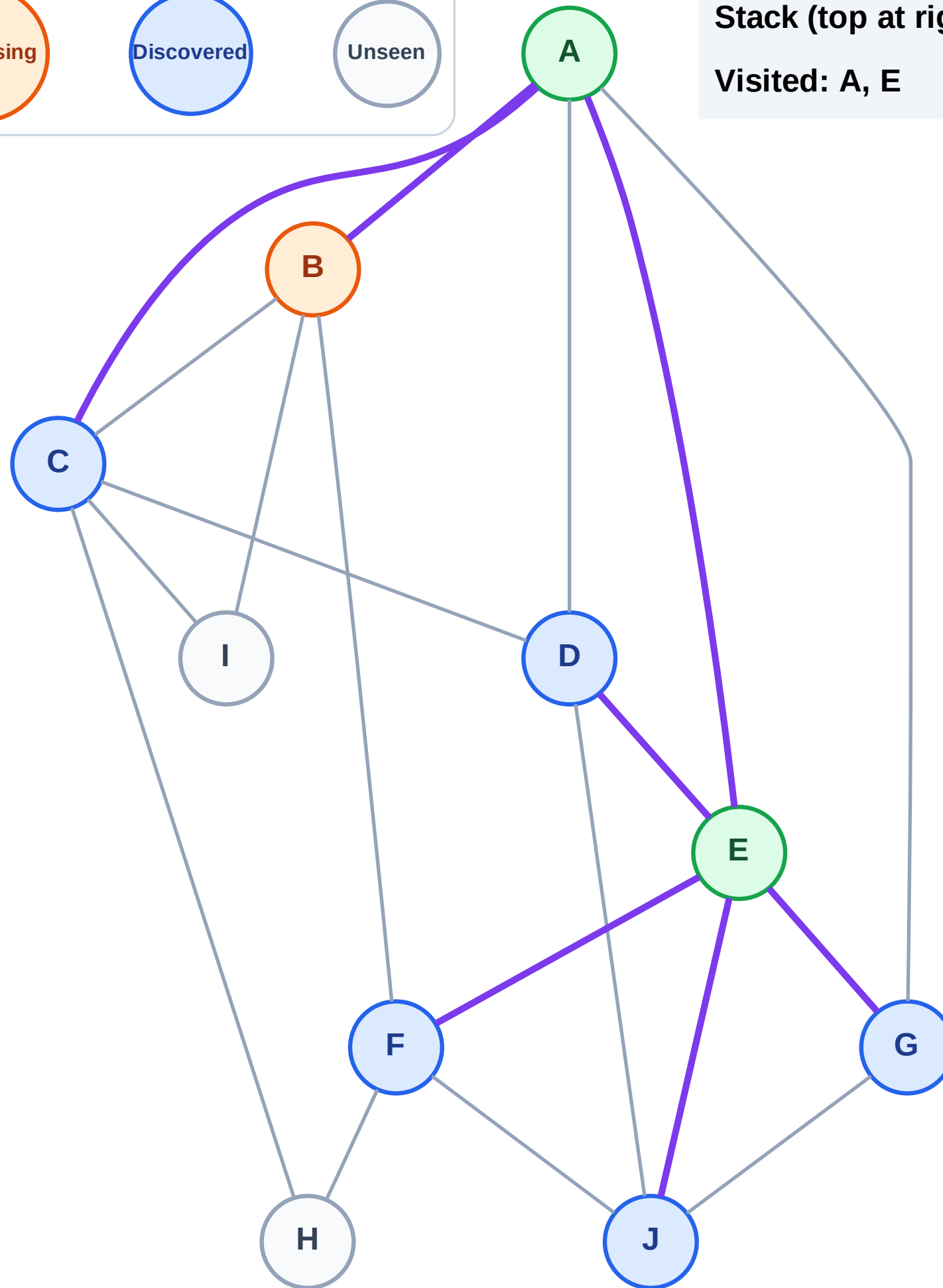
Step 6: Process node B

Legend



Stack (top at right): J | G | F | D | C

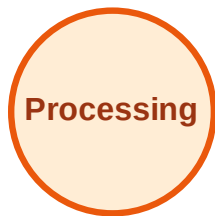
Visited: A, E



DFS Traversal

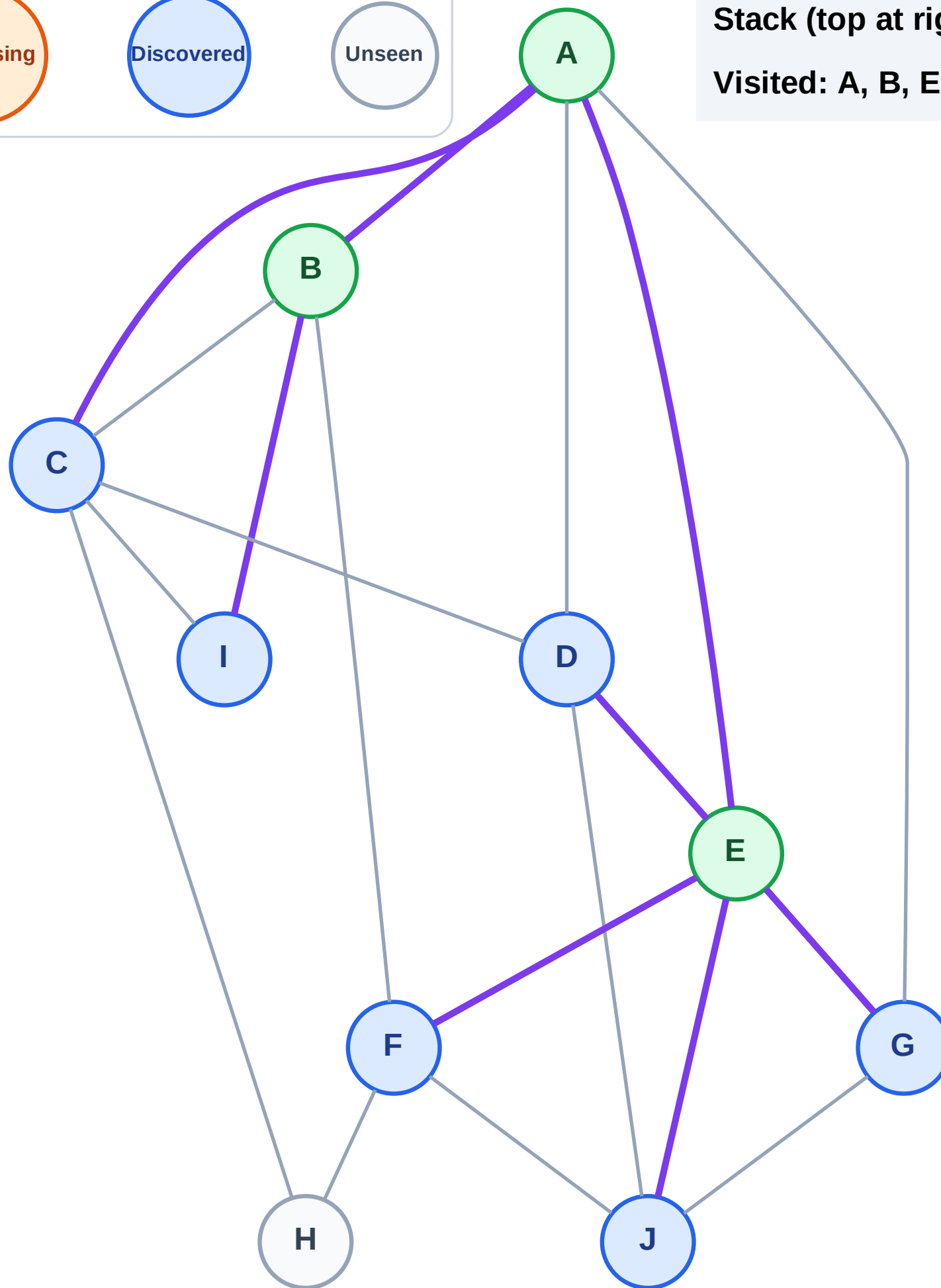
Step 7: Discover I from B

Legend



Stack (top at right): J | G | F | D | C | I

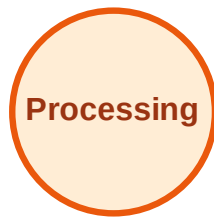
Visited: A, B, E



DFS Traversal

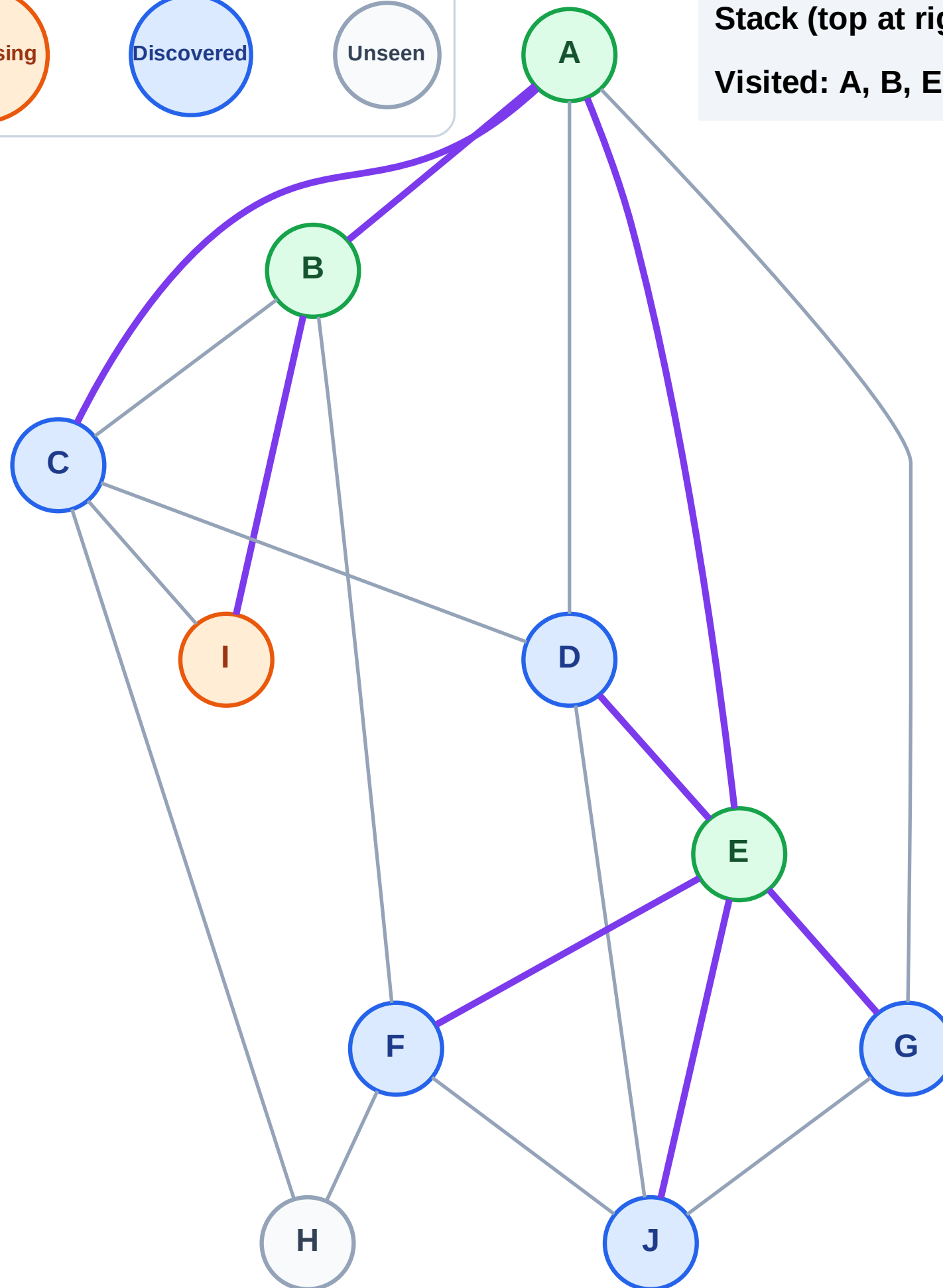
Step 8: Process node I

Legend



Stack (top at right): J | G | F | D | C

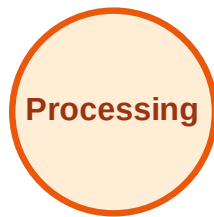
Visited: A, B, E



DFS Traversal

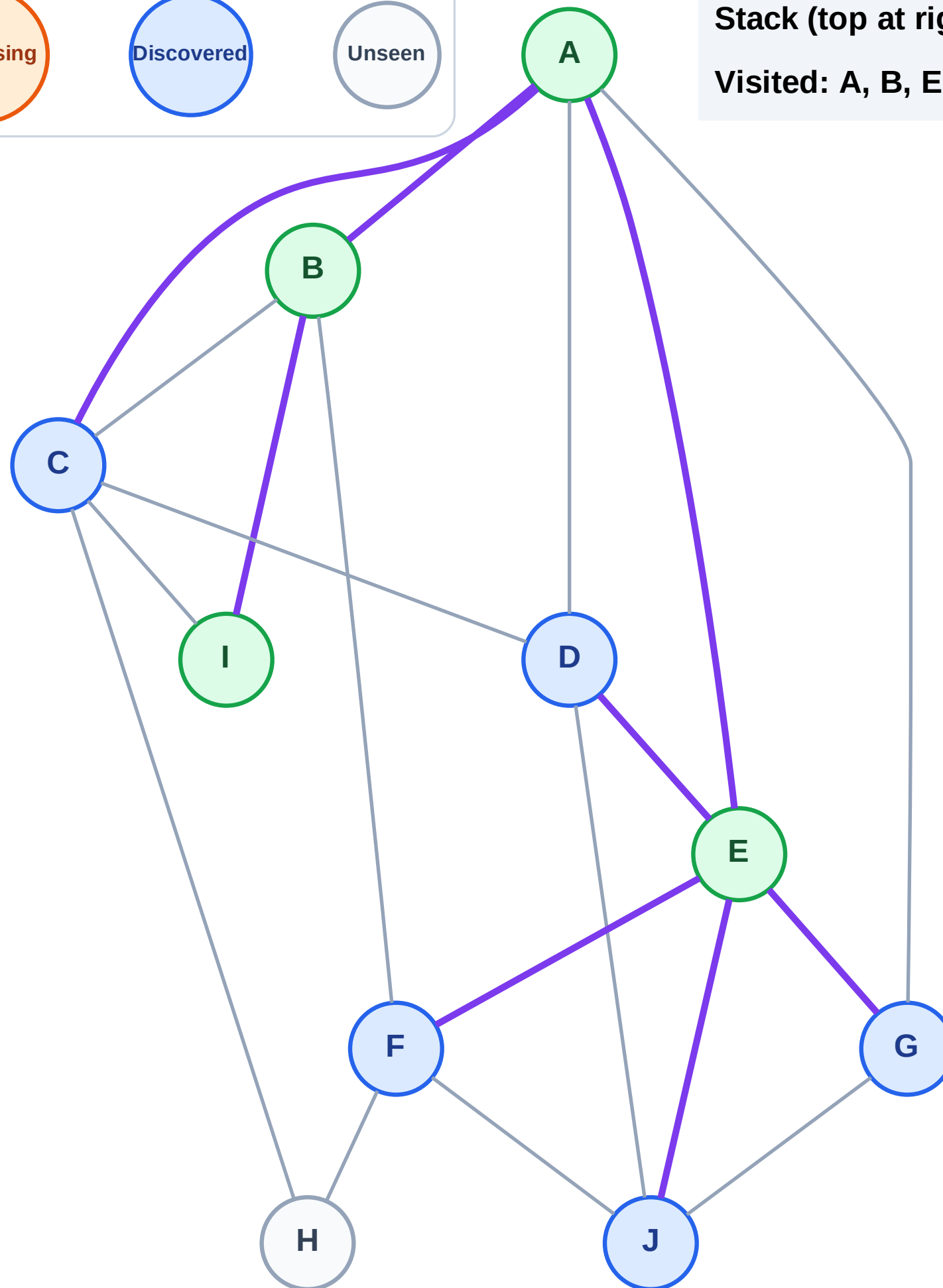
Step 9: No new neighbors from I

Legend



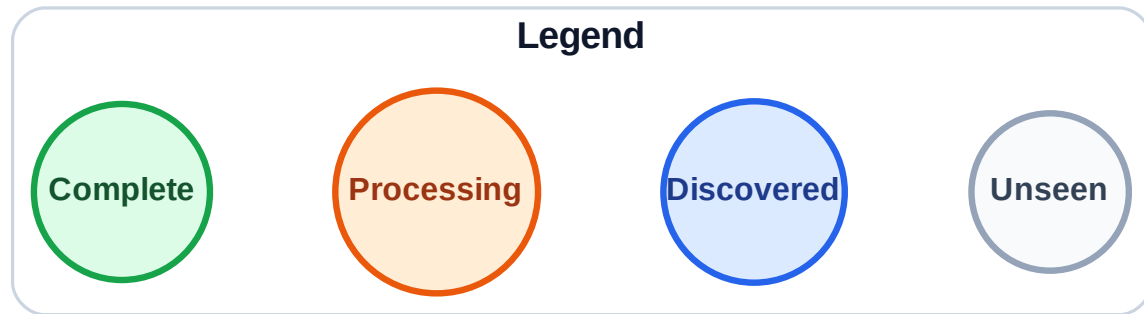
Stack (top at right): J | G | F | D | C

Visited: A, B, E, I

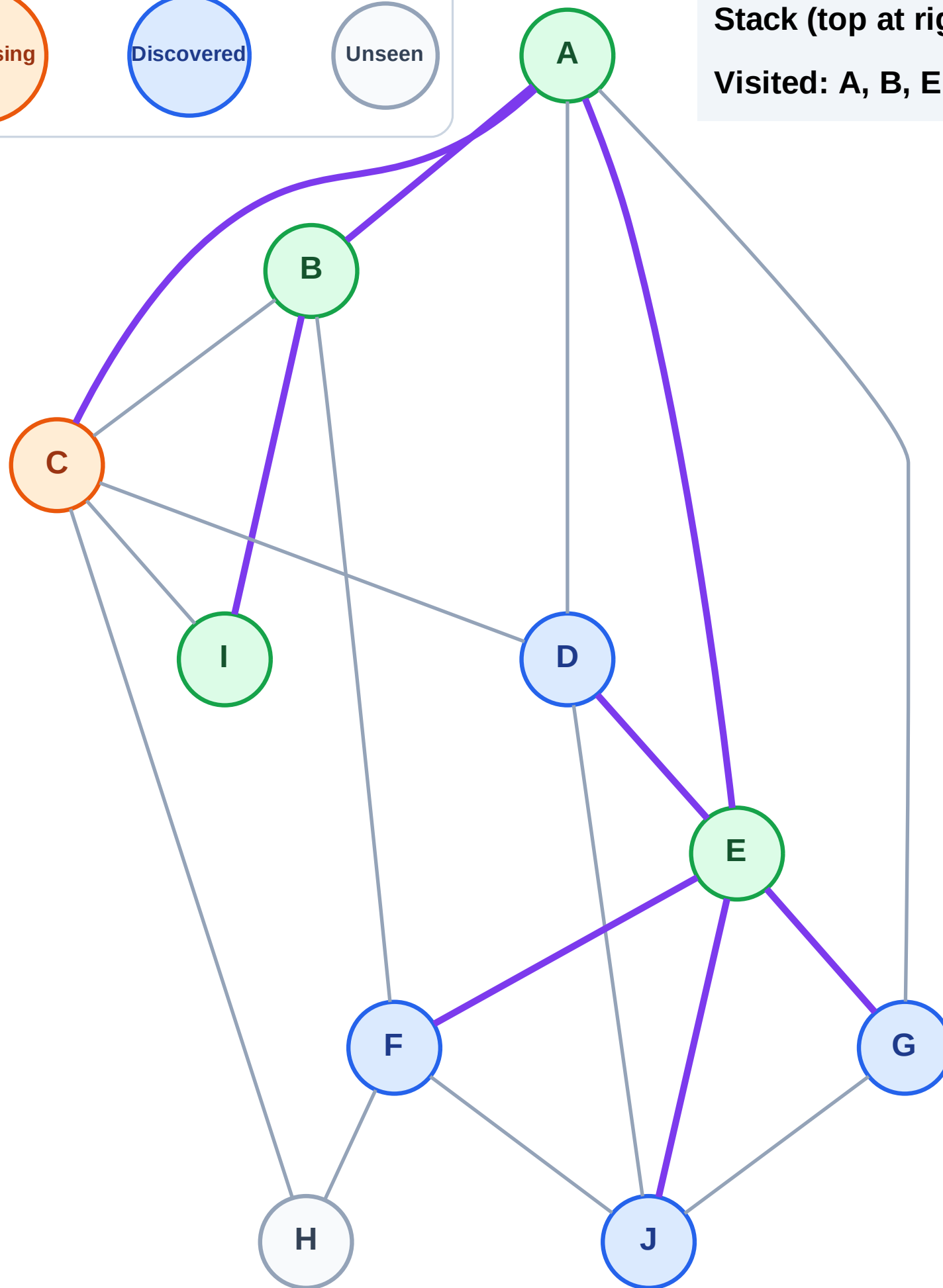


DFS Traversal

Step 10: Process node C



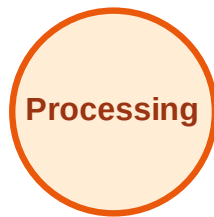
Stack (top at right): J | G | F | D
Visited: A, B, E, I



DFS Traversal

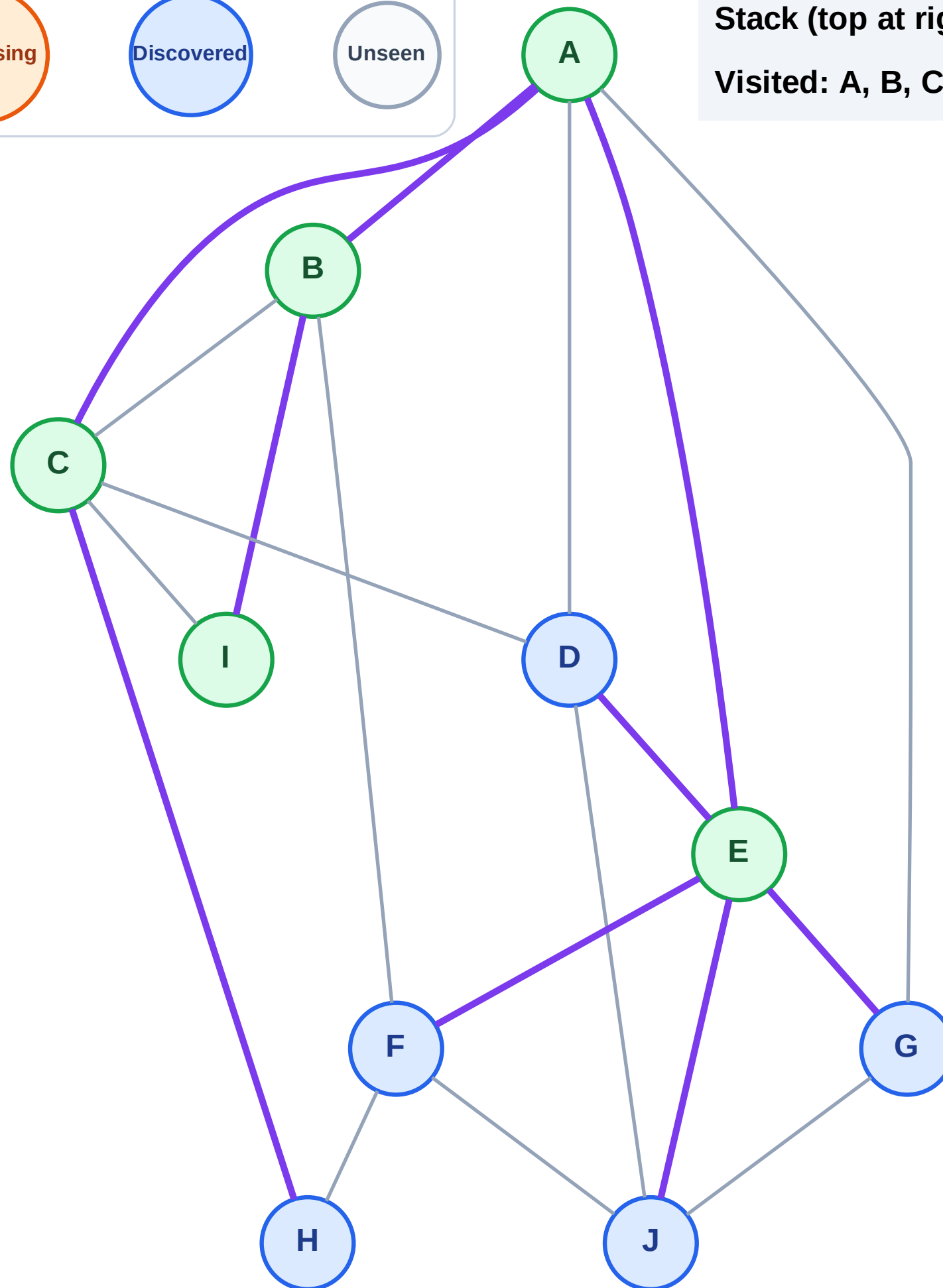
Step 11: Discover H from C

Legend



Stack (top at right): J | G | F | D | H

Visited: A, B, C, E, I



DFS Traversal

Step 12: Process node H

Legend



Complete



Processing



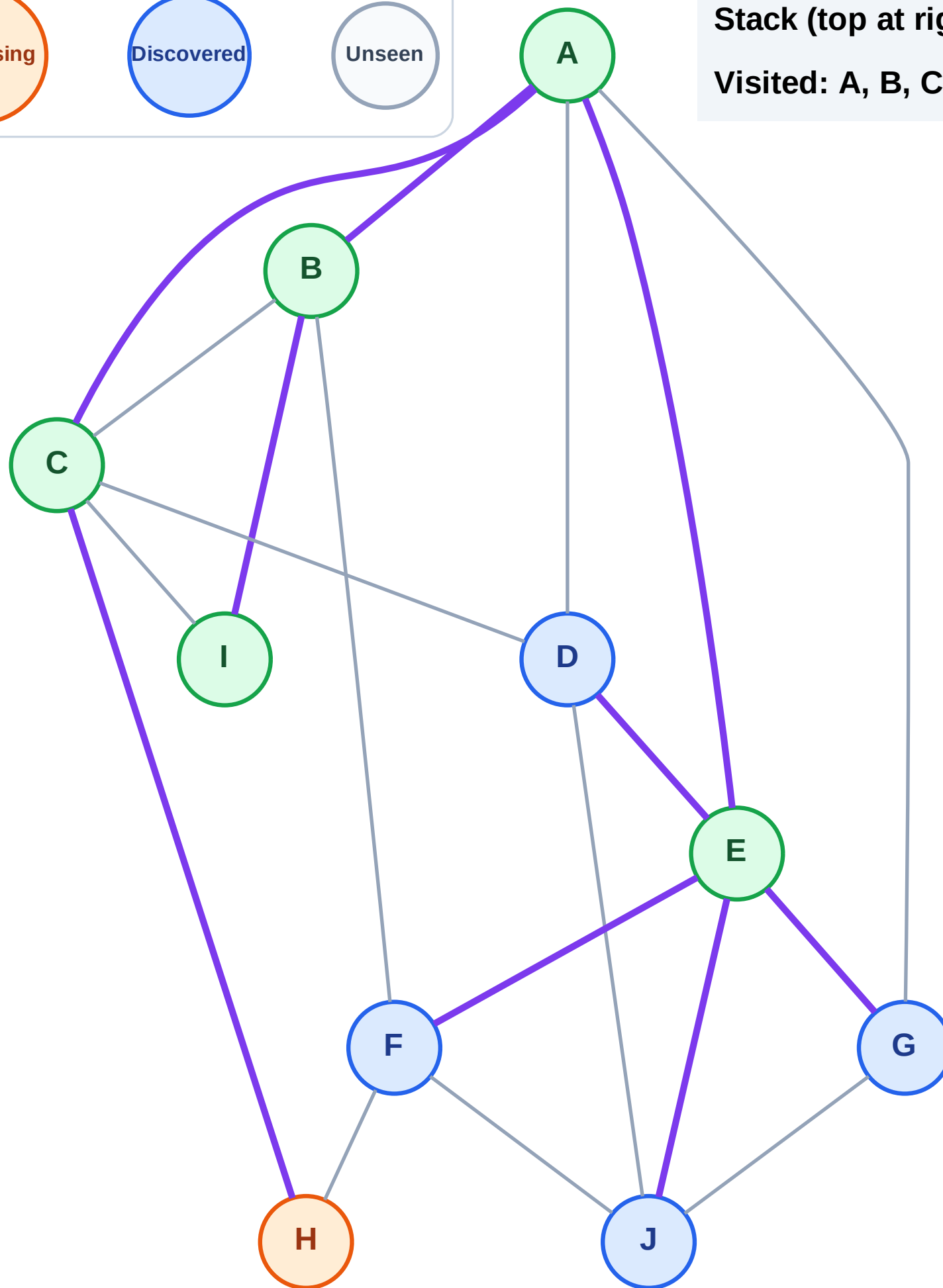
Discovered



Unseen

Stack (top at right): J | G | F | D

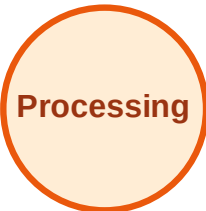
Visited: A, B, C, E, I



DFS Traversal

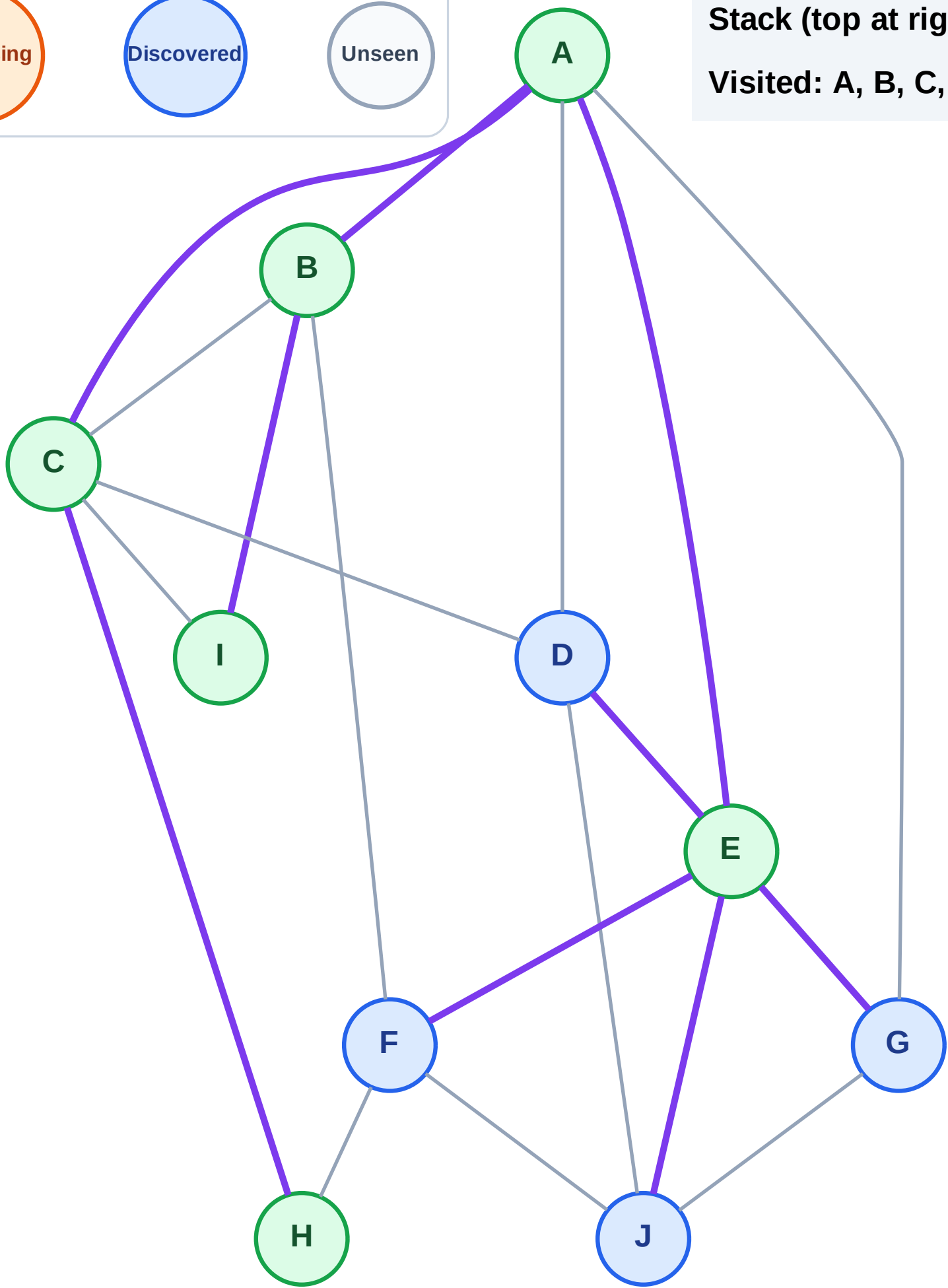
Step 13: No new neighbors from H

Legend



Stack (top at right): J | G | F | D

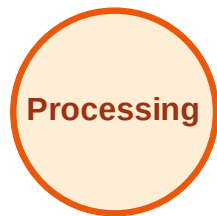
Visited: A, B, C, E, H, I



DFS Traversal

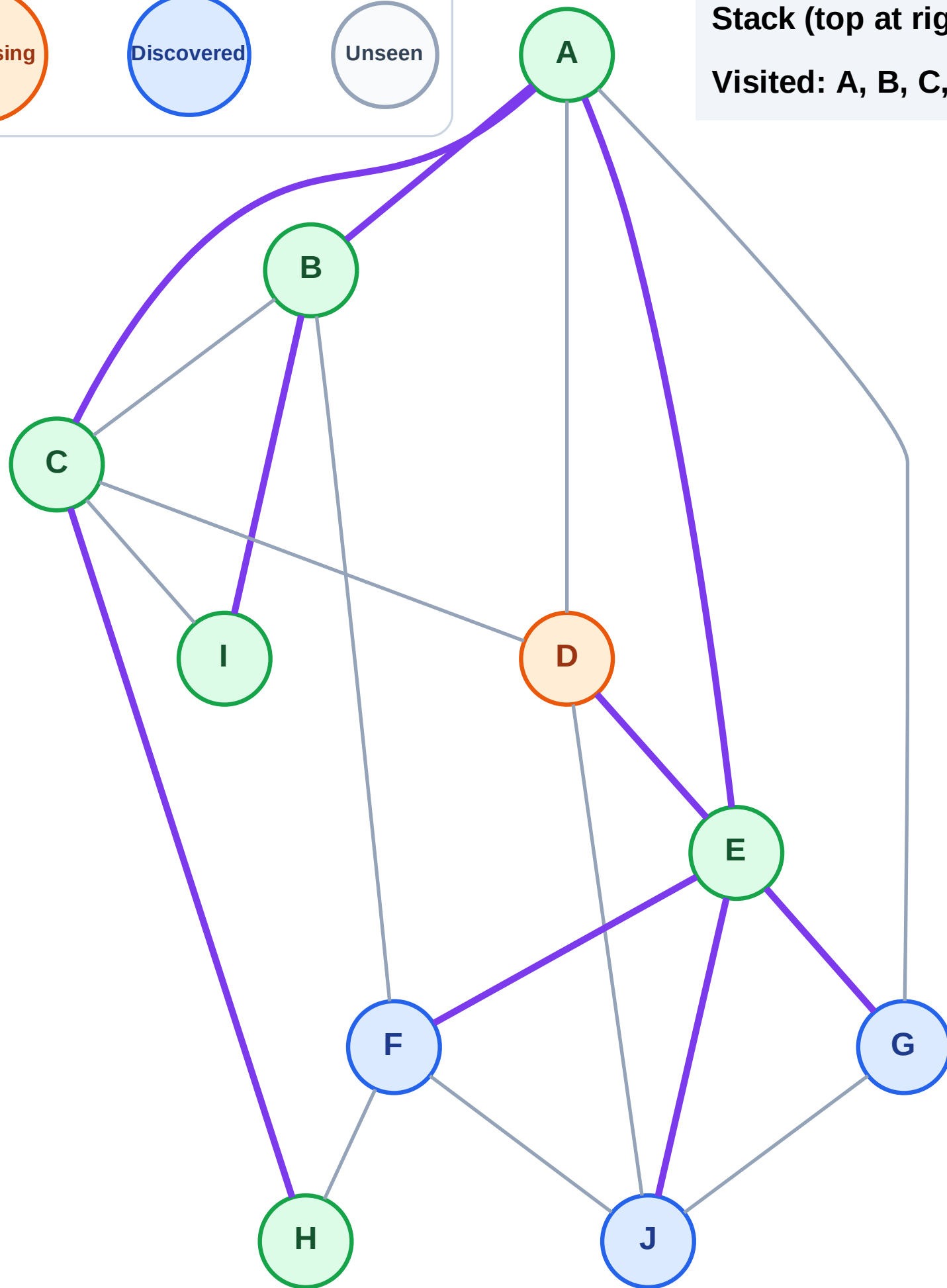
Step 14: Process node D

Legend



Stack (top at right): J | G | F

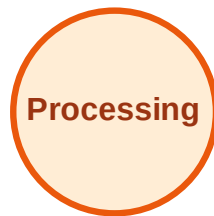
Visited: A, B, C, E, H, I



DFS Traversal

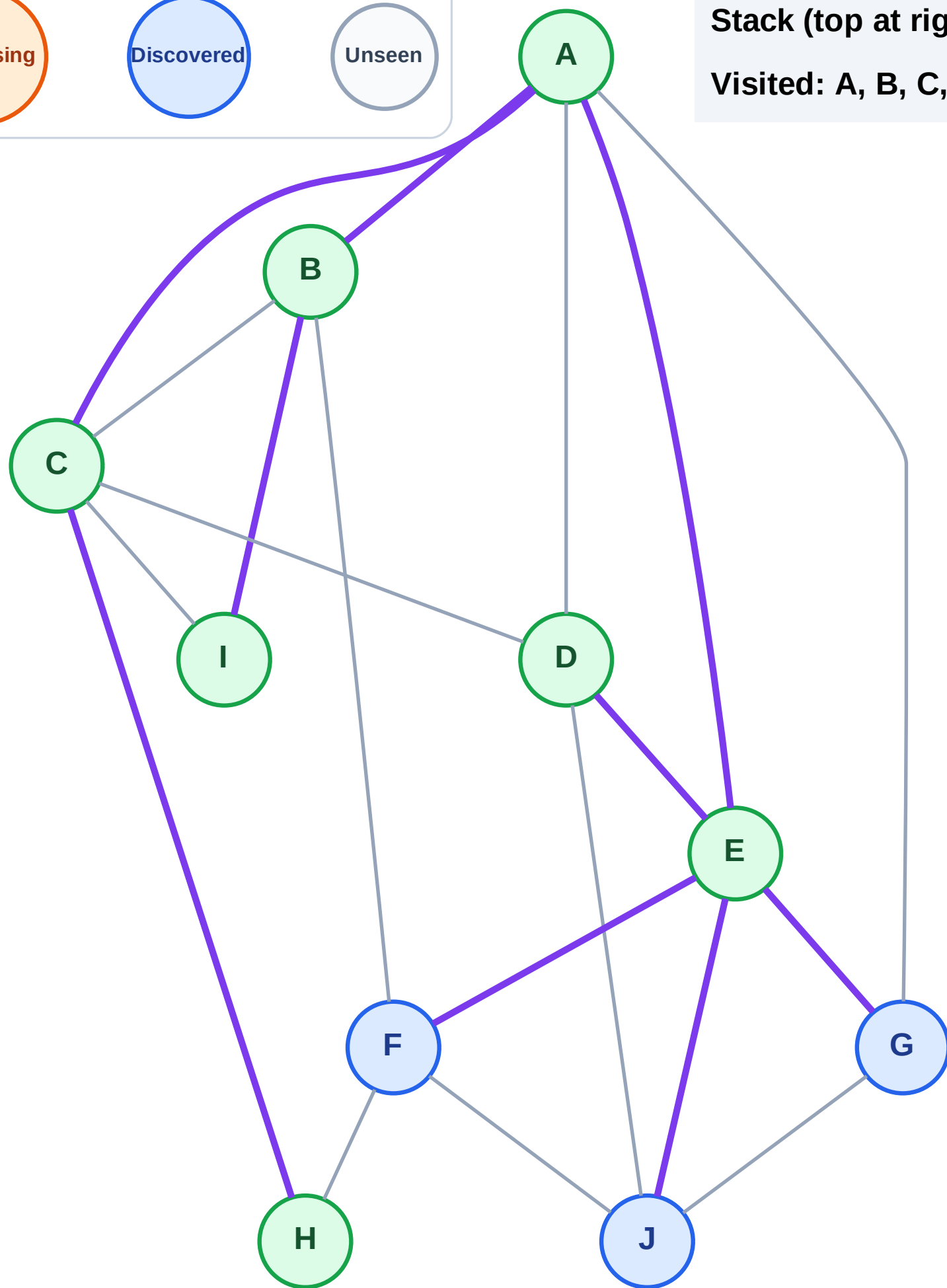
Step 15: No new neighbors from D

Legend



Stack (top at right): J | G | F

Visited: A, B, C, D, E, H, I



DFS Traversal

Step 16: Process node F

Legend

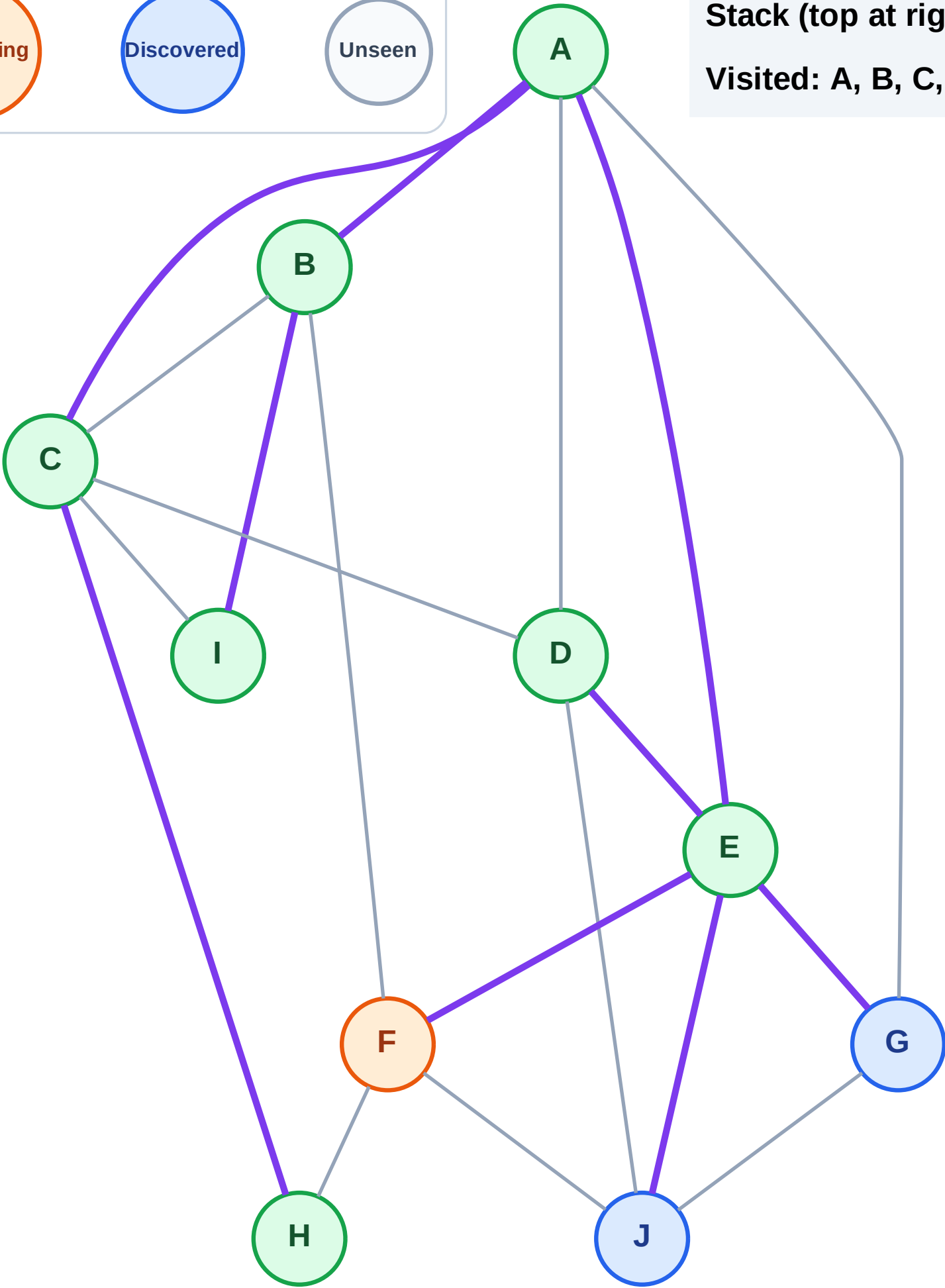
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G
Visited: A, B, C, D, E, H, I



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

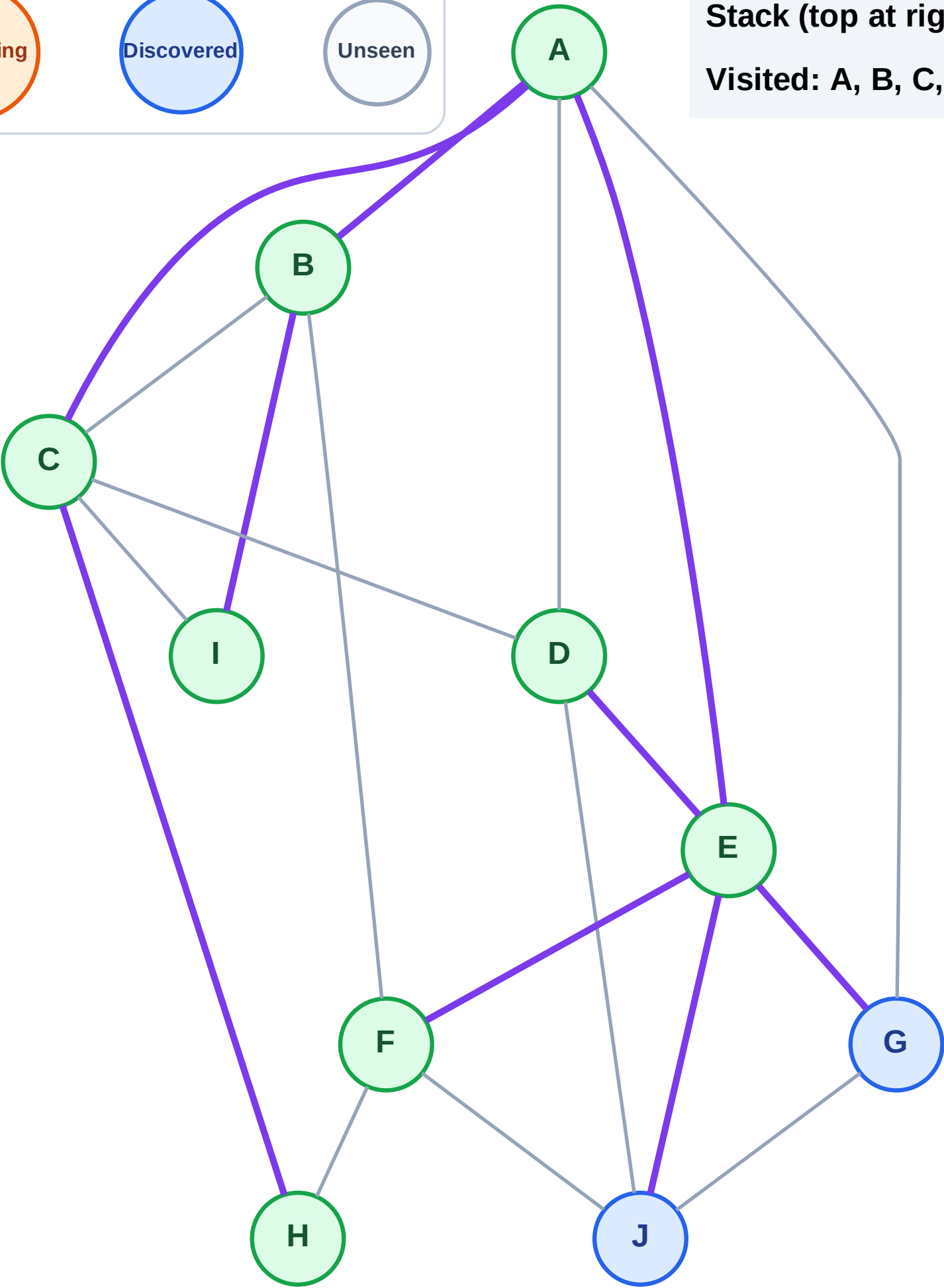
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node G

Legend

Complete

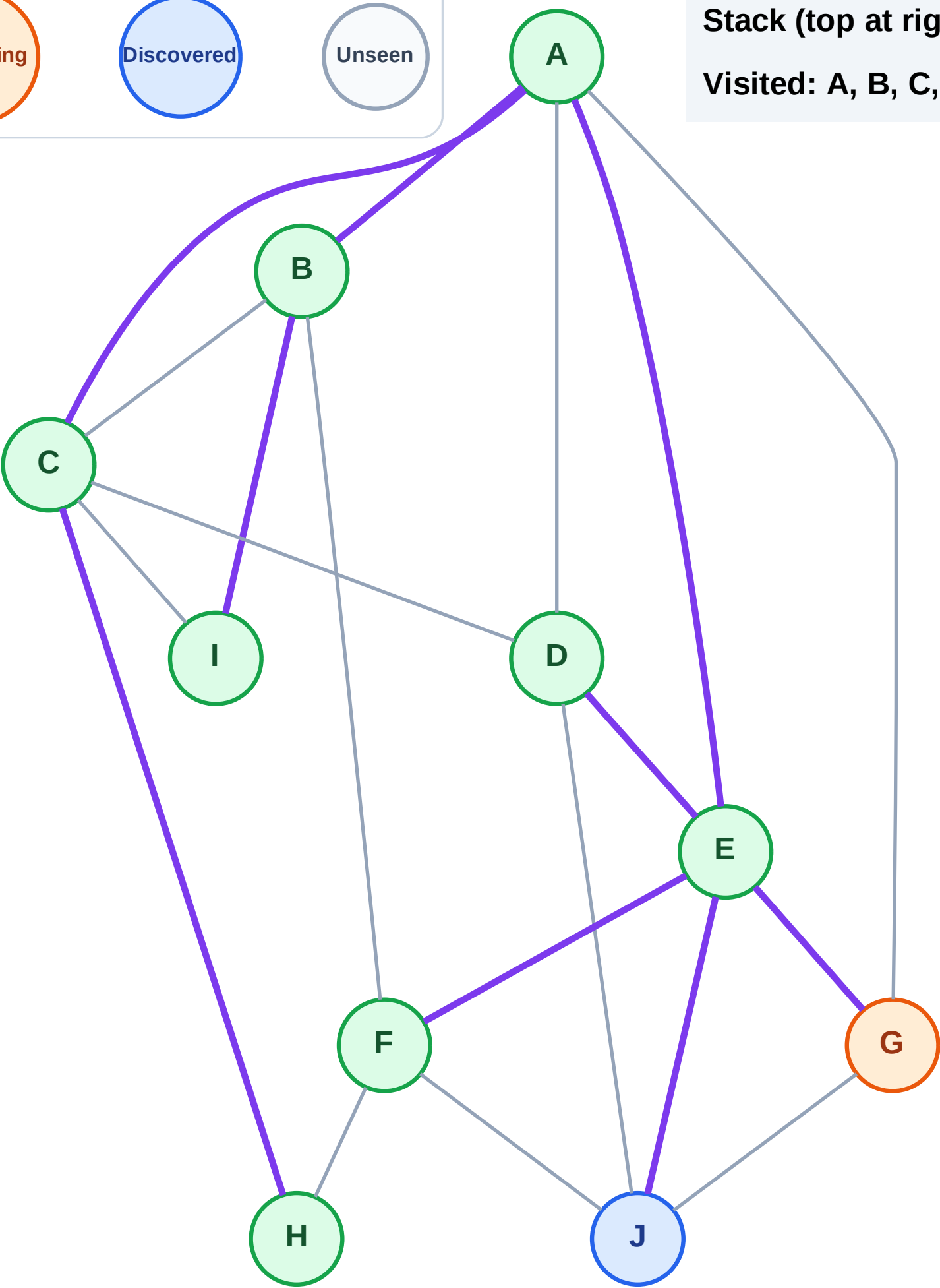
Processing

Discovered

Unseen

Stack (top at right): J

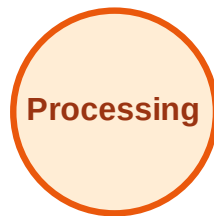
Visited: A, B, C, D, E, F, H, I



DFS Traversal

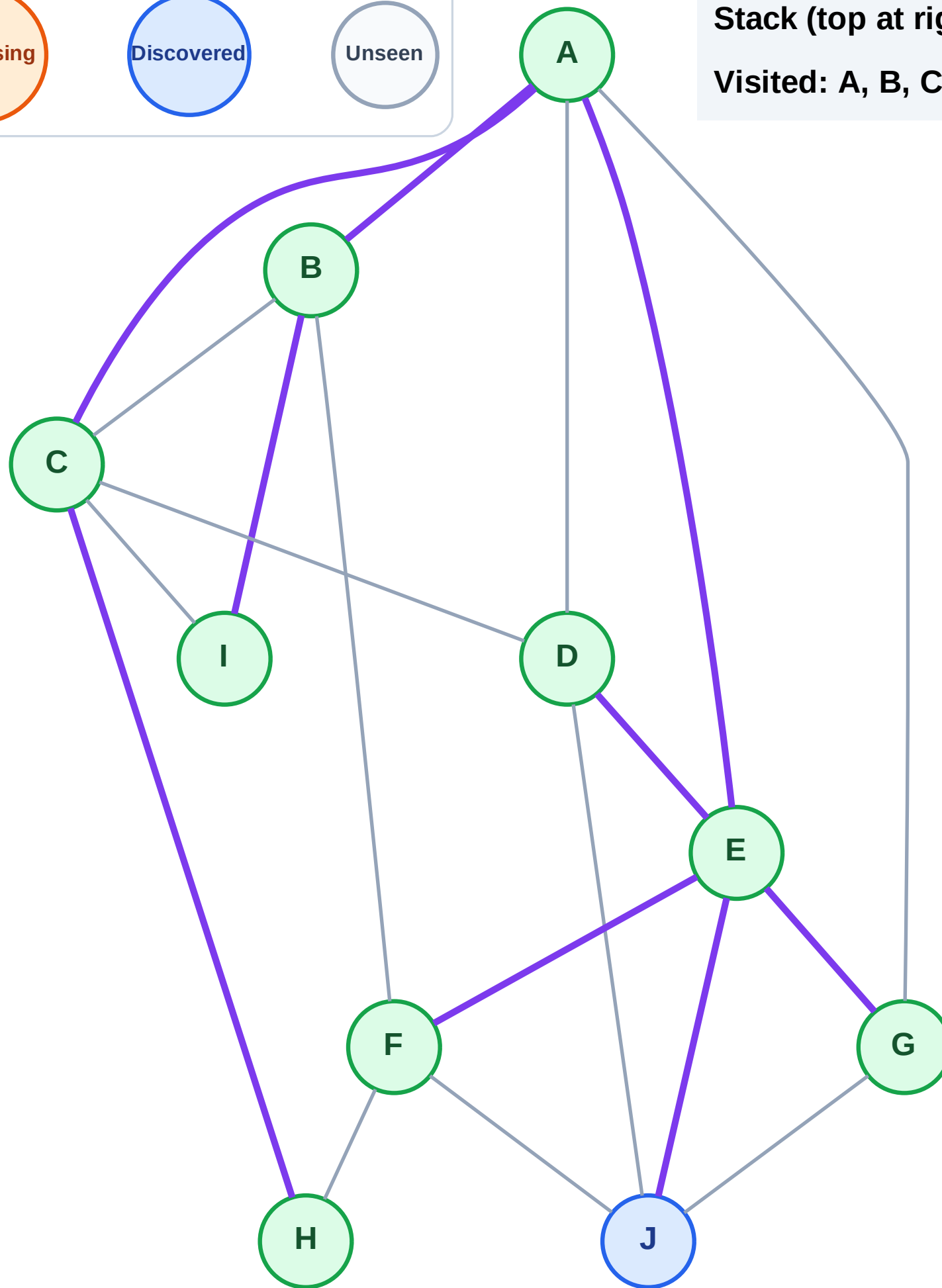
Step 19: No new neighbors from G

Legend



Stack (top at right): J

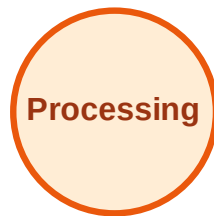
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

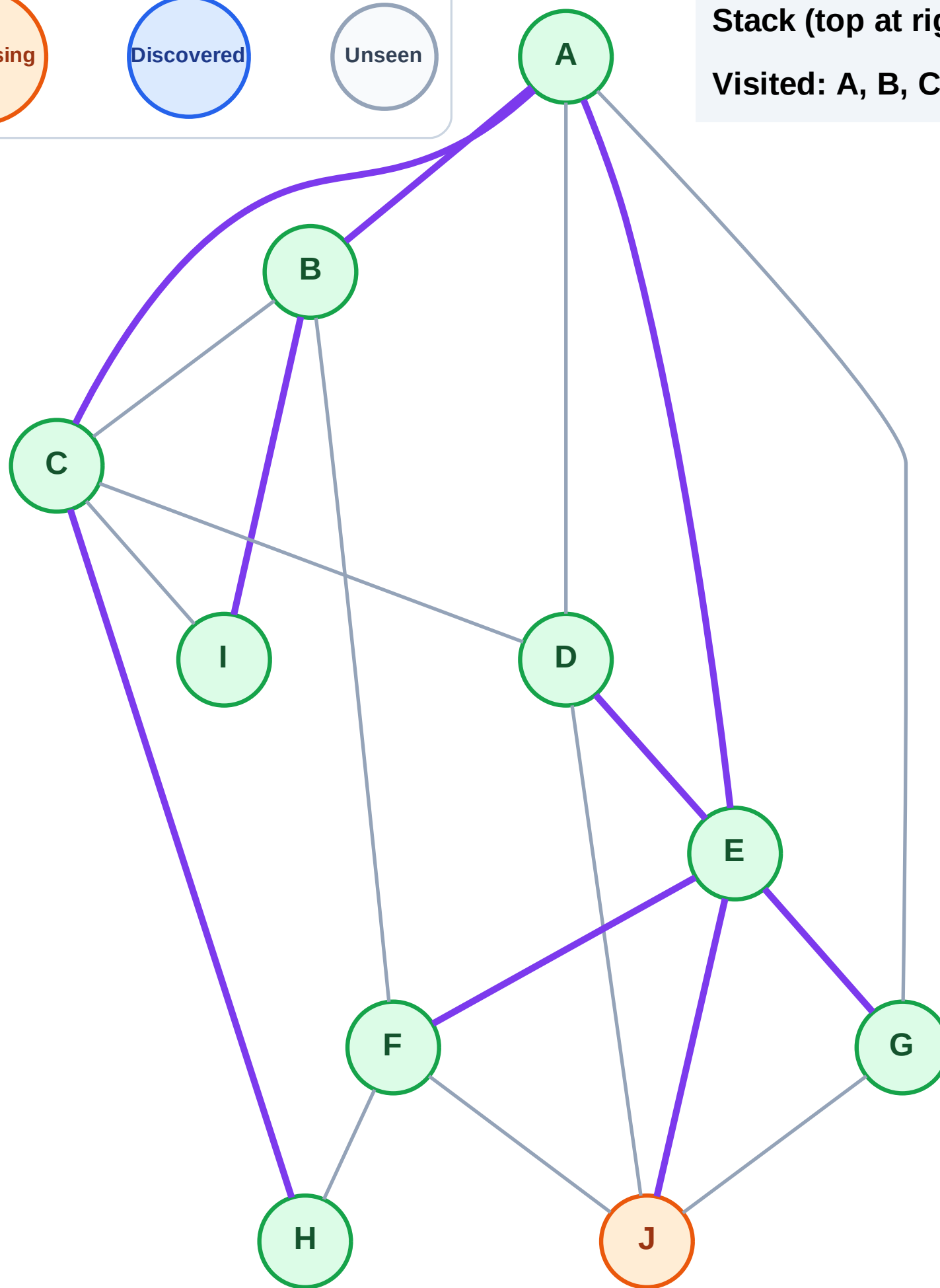
Step 20: Process node J

Legend



Stack (top at right): (empty)

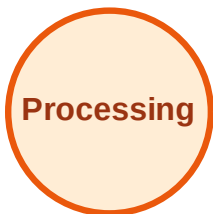
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

